

g·NAUTILUSPRO

WIRELESS BIOSIGNAL ACQUISITION

INSTRUCTIONS FOR USE



Instructions for Use
g.Nautilus PRO

Document version:
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Software version:

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UDI Sticker

CONTENTS

1	Important Notes.....	1
1.1	Warning and safety notice.....	1
1.2	Special treatment of electrodes and cables	2
1.3	Warranty/replacement of electrodes.....	2
1.4	Limitations of Reprocessing	2
1.5	Inspection.....	2
1.6	Interference	3
1.7	Intended use.....	3
1.8	Limitation	3
1.9	Intended user	3
1.10	The intended environment of use	3
1.11	Recommended electrodes and electrode gel	4
1.12	Properties of PC or notebook	4
1.13	Cybersecurty recommendations	4
1.14	2.4 GHz transmission	5
1.15	Prescription device	5
2	g.Nautilus PRO Introduction.....	6
2.1	Highlights.....	7
2.2	Different versions of g.Nautilus PRO	8
3	g.Nautilus PRO components.....	11
4	Switches and LEDs on the Headset	13
4.1	Indications of operation mode	13
4.2	Indications of charging mode.....	14
5	Markings on the top side of the g.Nautilus Headset	15
6	g.Nautilus PRO Base Station, LED and USB cable	19
6.1	Indications of operation mode	21
6.2	Labels on the cable	21
7	Markings on the top side of the g.Nautilus Base Station	22
8	Charging device	26
8.1	Charging pad and USB cable	26
8.1.1	Labels on the charging pad	26
8.1.2	Labels on the USB cable.....	27
8.2	Power Supply for charging pad.....	28
8.2.1	Labels on the power supply.....	28
9	Packaging	30

10	Safe operation of g.Nautilus.....	31
10.1	Set up and measurement with the device	31
10.2	Patient environment.....	31
10.3	Avoiding electrostatic discharge.....	32
10.4	g.Nautilus PRO: Gel electrodes.....	32
10.5	g.Nautilus PRO: Dry electrodes	34
10.6	Avoiding electrode and cable movements.....	36
10.7	Avoiding electrostatic charges	36
10.8	Grounding	36
10.9	Switch off and store the device.....	37
10.10	Charging of g.Nautilus.....	38
10.11	Details for the correct maintenance of the accumulator.....	39
10.12	Cleaning and Disinfection of Headset, cap and electrodes	40
11	Initial instructions on how to make the device patient ready	41
11.1	Removing gel	41
11.2	Containment and Transport.....	41
11.3	Cleaning g.Nautilus PRO: Gel electrodes and cap	41
11.4	Disinfection of g.Nautilus PRO: Gel electrodes and cap	43
11.5	Cleaning g.Nautilus PRO: Dry electrodes and cap	44
11.6	Disinfection g.Nautilus PRO: Dry electrodes and cap	45
11.7	Cleaning and Disinfection of enclosure parts of Headset, Base Station and charging pad	46
11.8	How to change or replace electrode caps of different sizes (S/M/L) on a g.Nautilus PRO system.....	47
12	General notes.....	49
12.1	Classification.....	49
12.2	Transportation and storage conditions.....	49
12.3	Location details.....	49
12.4	Waste disposal details	49
12.5	Warranty.....	49
13	Declaration of conformity	50
14	Technical specifications.....	51
14.1	Amplifier Settings	52
14.2	Acceleration sensor	52
14.3	Analog-Digital-Converter (ADC)	52
14.4	Analog signal processing time and digital inputs	52
14.5	RF module	53
14.6	Maximum voltages at the following sockets on Base Station.....	53
14.7	Digital inputs	53

g·NAUTILUSPRO

15	Electromagnetic compatibility g.Nautilus PRO	54
15.1	Guidance and manufacturer's declaration – electromagnetic emission	54
15.2	Guidance and manufacturer's declaration – electromagnetic immunity	55
15.3	Guidance and manufacturer's declaration – electromagnetic immunity	56
15.4	Recommended separation distances between portable and mobile RF communications equipment and the g.Nautilus PRO	57
16	Appendix	59
16.1	Acceleration sensor	59
16.2	g.Nautilus PRO (32-channel setup)	60
16.3	g.Nautilus PRO (19-channel setup)	61
16.4	g.Nautilus PRO (16-channel setup)	62
16.5	g.Nautilus PRO (8-channel setup)	63
16.6	g.Nautilus PRO (32-channel setup CSP system)	64
16.7	g.Nautilus PRO (16-channel setup CSPmB system)	65
16.8	g.Nautilus PRO (16-channel setup CSPrX system)	66
16.9	g.Nautilus PRO Base Station pin assignment for digital inputs	67
16.10	Information to aid in the detection of failures	68
16.11	Manual Best Channel Selection	70

1 Important Notes

In the following document, the term g.Nautilus refers to both g.Nautilus PRO wet and dry systems. Depending upon your purchased device, the specific paragraphs for the wet or dry system apply. Please check your purchasing documentation or the device label to identify your device type.



ATTENTION

- Conductible parts of all electrodes must not have contact with the earth or other conductible parts
- Avoid electrostatic discharge impulses when handling g.Nautilus or touching the electrodes
- All components of the device must not be used in humans with pace-makers or electrical stimulators
- Pay attention to the precautions regarding electromagnetic compatibility
- The operation of the device can be compromised within shielded rooms. In case of problems, relocate the receiving device or consult g.tec Technical Support (see page II how to contact g.tec).
- Recordings with more than 1 device with 500 Hz in the same location could lead to increased data loss and are therefore not recommended
- The operator has to be familiar with the operation of g.Nautilus and must operate the device according to the instructions for use
- The device and its accessories must not be exposed to increased mechanical stress
- Each time you use g.Nautilus, you must first check the device and its accessories for possible damage to connectors, sockets and cables. Check the electrode cable connections with special care and ensure that the electrode lines have no breaks or cracks. Any cables, connectors, accessories, or other parts of the equipment must be replaced immediately if damaged or not working correctly.
- The Headset is not protected against electrical defibrillation - before defibrillation, the electrodes must be removed from the subject!
- Only use accessories identified for use with this device
- If g.Nautilus is used at the same time as other devices on subjects, the following leakage currents have to be checked:
 - Ground leakage current
 - Enclosure leakage current
 - Patient leakage current

Furthermore, all configurations shall comply with the system standard IEC 60601-1. Everybody who connects additional equipment to the signal input part or signal output part configures a medical system, and is therefore responsible for ensuring the system complies with the requirements of the system standard IEC 60601-1. If in doubt, consult the technical service department or your local representative.

1.1 Warning and safety notice

The device is powered internally via a Lithium-Ion accumulator. The accumulator must only be replaced by the manufacturer. Opening the device destroys the housing. Hence, the device must be sent back to the manufacturer for any visual inspection of the inside of g.Nautilus.

1.2 Special treatment of electrodes and cables

g.tec uses special lightweight, thin and highly flexible cables for active electrodes to provide high comfort and easy cap mounting, especially for multi-channel recording. These cables are sensitive and need to be treated with special care. Following some basic guidelines will prolong the lifetime of electrodes and cables:

- never pull on electrode cables
- avoid knots in cables
- do not soak cables and electrodes longer than necessary
- avoid exposure to direct sunlight or chemical agents
- do not autoclave electrodes
- make sure that no gel remains on electrodes or cables after cleaning
- protect from contamination with gel, water or disinfectant
- always make sure that electrodes, cables and caps are completely dry before storing
- don't cut or pinch electrode cables; light bending is safe

1.3 Warranty/replacement of electrodes

Electrode and cable lifetime depends on proper usage, careful treatment and cleaning, and appropriate storage. g.tec will provide warranty replacement only if there is no visible physical damage to the parts, such as: damaged, broken or pinched cables; isolated, eroded contact pellets; or damaged housings or connectors.

1.4 Limitations of Reprocessing

- do not use any other detergent as mentioned in this policy!
- do not perform automated reprocessing in Washer Disinfectors (WD) or Endoscope Washer Disinfectors (EWD)!
- do not machine-wash!
- do not use a laundry dryer or other hot air devices!
- do not put into an ultrasonic bath!
- do not autoclave caps or electrodes!

For disinfection (if required) use 'Anioxyme XL3', 'Sekusept Plus' or Metricide® cold sterilizing solution or disinfection alcohol like isopropyl alcohol (70 %, 5 minutes) only. Do not exceed the treatment time indicated in the Aniosyme, Sekusept or Metricide manuals. 'Aniosyme XL3' is a product of ECOLAB (www.ecolab.com) and is available from most local healthcare providers.

1.5 Inspection

The manufacturer is responsible for the safety, performance and reliability of the device as supplied to the customer at the time of delivery. This responsibility expires if the device is changed. Please note the following:

- changes to the device must be performed by the manufacturer only, and service and repair is performed by corresponding qualified personnel only
- the device must be used according to the instructions for use

According to EN 62353:2014, the device and its accessories must be checked once every two years (minimum).

1.6 Interference

g.Nautilus and its components have been tested and comply with the electromagnetic compliance limits for the Medical Device Directive 93/42/EEG (EN 60601-1-2:2007 Class B). See the chapter on Electromagnetic compatibility g.Nautilus PRO. The equipment, if not installed and used in accordance with the instructions, may cause interference with other devices in the vicinity. If this equipment does interfere with other devices, which can be determined by turning the equipment off and on, try to correct the interference through one or more of the following measures:

- reorient or relocate the receiving device.
- increase the separation between the equipment.
- consult g.tec Technical Support (see page II how to contact g.tec).

1.7 Intended use

The g.Nautilus PRO is intended to be used to acquire the electroencephalogram (EEG) and transmit it wirelessly to a computer.

1.8 Limitation

The device **must not** be used for patient monitoring. The device **must not** be used for the determination of brain death. Additional examinations are needed for diagnosis, and no diagnosis may be done based only on using this device.

The device is not suitable for high density recordings for source localization, diagnostic investigations related to brain diseases as in routine clinical settings, combined EEG with fMRI¹, invasive recordings, recordings on non-intact skin and EEG recordings where very high sample rates are necessary (BAEPs²).

As g.Nautilus PRO provides 8 independent digital inputs, for event related experiments the number of events, which could happen at the same time, is limited to 8. Limitation regarding the electrode grid layouts is addressed in section "[Grid limitation](#)" of "[Different Versions of g.Nautilus PRO](#)".

1.9 Intended user

Researcher or medical doctor with experience in EEG measurements.

1.10 The intended environment of use

The device **must not** be used in dangerous conditions such as wet rooms or explosive environments. The relative humidity must be between 25 % and 80 %. The device **must not** be used in combination with any other high-frequency device. Using a high frequency device with g.Nautilus PRO can cause burning under the electrodes and could damage the device. Please consider the environmental conditions for safe operation, storage and transportation described in Section 12.

¹ functional magnetic resonance imaging

² brainstem auditory evoked potentials

1.11 Recommended electrodes and electrode gel

g.Nautilus PRO (wet system) is delivered with a fixed, non-exchangeable electrode configuration pre-mounted in an electrode cap (maximum 32 electrodes in defined arrangement with reference- and ground-electrode). For this system, use electrode gel (model: E9) from Electro-cap Intl., Inc. (K111717).

g.Nautilus PRO (dry system) does not require electrode gel. It is delivered with a fixed, non-exchangeable electrode configuration pre-mounted in an electrode cap (maximum 32 electrodes in defined arrangement with reference- and ground- clip). For the ground and reference electrodes, use adhesive disposable Ag/AgCl electrodes (recommended: Kendall H135SG from Tyco Healthcare (K953649), or Kendall H135T from Cardinal Health (22B1X00007DG001A)). For the EEG channels, use only the gold coated dry electrodes included in delivery for connection to the electrode clips.

1.12 Properties of PC or notebook

g.Nautilus requires a PC, notebook or embedded computer with UPS³ (certified according to the respective IEC standards (e.g. IEC 60950 or IEC 62368-1 for data processing equipment, respectively)) running a Microsoft Windows operating system (Windows 11 Professional English with 64 Bit). The following table shows the minimum requirements:

Hardware	Minimum Requirements
CPU	1 gigahertz (GHz) or faster with 2 or more cores on a compatible 64-bit (x64) processor
Hard disk	64 GB
RAM	4 GB
USB 2.0 port (EHCI – enhanced host controller interface)	One free USB port for each g.Nautilus Base Station
Uninterruptible power supply (UPS)	

1.13 Cybersecurity recommendations

- Make sure that developed software is deployed as compiled code with obfuscation if applicable.
- Deploy strong code integrity policies to allow only authorized apps to run.
- Use endpoint detection and response solutions that can automatically detect and remediate suspicious activities.
- Ensure a highly secure build and update infrastructure by promptly installing security updates for operating systems and software, enforcing mandatory integrity checks to allow only trusted tools, and requiring multi-factor authentication for administrators.
- Ensure that all update channels are secured with SSL and implement certificate transparency to validate the legitimacy of issued certificates.
- Ensure that all elements - such as configuration files, XML files, and packages - are digitally signed to maintain integrity and authenticity.
- Establish an incident response process, transparently disclose supply chain incidents, and promptly notify customers with accurate information.

³ uninterruptible power supply

- Make sure that your hard disks are encrypted using e.g. BitLocker.
- Change your passwords in regular intervals.
- Make sure to have a physical access control for your environment.
- Take care of phishing and spoofing attacks.
- Make sure that qualified personnel/users are trained correctly in their executive function.
- Consider that sensitive data must not be stored in plain text, environment variables, or hard-coded anywhere.
- Security-relevant constants must not be hard-coded.
- All relevant data must be logged to a file. This log file must not contain sensitive data.

1.14 2.4 GHz transmission

g.Nautilus uses the 2.4 GHz band for wireless transmission. Ensure that enough transmission bandwidth is available in your environment, since other devices might also use the same band (e.g. WiFi or Bluetooth devices). Use wireless screening tools to ensure the availability of the necessary transmission channel.

1.15 Prescription device

Caution: US federal law restricts the devices described herein to sale by or on the order of a physician.

2 g.Nautilus PRO Introduction

g.Nautilus PRO is g.tec's biopotential amplifier with wireless data transmission technology with active wet or dry electrodes (8-32 channels). The device can acquire EEG data with 24 Bit resolution and a sampling rate of 250 or 500 Hz. It also has a 3-axis acceleration sensor that can sense ± 6 g.

Up to 32 analog to digital converters perform the simultaneous sampling. Each analog to digital converter operates at 1.024 MHz. A corresponding down-sampling then yields a user-selectable sampling rate of 250 or 500 Hz. A sampling rate of 250 Hz means an oversampling of 4096, yielding a high signal to noise ratio.

The device is equipped with an internal impedance check to determine the electrode-skin-impedance. g.Nautilus is controlled via an Application Programming Interface (C-API).

A Base Station receives the digitized EEG data and can be connected to any free USB port of the PC or notebook.

g.Nautilus PRO with 32 active wet electrodes





The Headset of the biosignal amplifier g.Nautilus with wireless transmission is attached to the back side of the cap.

g.Nautilus PRO is equipped with gel-based active electrodes (LADYbird technology), or with dry active electrodes (SAHARA technology). The electrodes (LADYbird) or electrode clip connectors (SAHARA) are connected to the g.Nautilus Headset with special flexible PCB strands instead of wires and cannot be disconnected from the device Headset. Thus, the electrode system can be mounted very quickly, and this approach avoids mistakes caused when connecting the electrodes' wires to each electrode.

2.1 Highlights

- wireless EEG recordings with high signal quality
- very lightweight to minimize artifacts
- active wet or active dry electrodes for fast mounting without using abrasive gel
- up to 32 analog input channels with 24 Bit resolution
- sampling rate of 250 or 500 Hz per channel
- digital filtering
- oversampling to achieve a high signal to noise ratio

- g.Nautilus Base Station can be connected to a computer
- simultaneous sample and hold for all channels
- easy and fast application of electrodes via prefixed positions
- wireless charging of the accumulator
- adjustable input sensitivity of ± 187.5 mV up to ± 2.25 V
- three axis acceleration sensor

The following table shows the available input ranges and the corresponding voltage resolution:

Input range (mV)	Voltage resolution (nV)
± 187.5	22.35
± 375	44.70
± 562.5	67.06
± 750	89.41
± 1125	134.11
± 2250	268.22

2.2 Different versions of g.Nautilus PRO

g.Nautilus PRO is available with several different electrode layouts, ranging from 8 to 32 channels with one ground and one reference channel. The additional reference channel allows the input stage to work in fully-differential mode. This brings the advantage of maximum noise rejection, an increased dynamic range and increased SNR. The appendix shows the different electrode configurations in detail. The following electrode layouts are available:

32 electrodes distributed according to the extended 10/20 electrode system

- Covered regions: frontal, central, temporal, parietal
- Advanced BCI experiments (P300, motor imagery, visual evoked potential (VEP), steady state visually evoked potential (SSVEP), etc.)
- Advanced psycho-physiological experiments (event-related potential (ERP))

19 electrodes distributed according to the extended 10/20 electrode system

- Covered regions: frontal, central, temporal, parietal
- Neurophysiological experiments
- Simple psycho-physiological experiments (event related potentials)

16 electrodes distributed according to the extended 10/20 electrode system

- Covered regions: frontal, central, temporal, parietal
- BCI experiments
- Simple psycho-physiological experiments (event related potentials)

8 electrodes distributed for performing P300 experiments

- Covered regions: parietal
- BCI experiments (P300 spelling)

32 electrodes with CSP layout distributed for performing motor imagery based brain-computer interface experiments

- Covered regions: frontal, central, temporal
- Advanced BCI experiments (motor imagery with common spatial patterns, source derivations over the sensory motor cortex)
- Advanced psycho-physiological experiments focused on sensory motor cortex activation

16 electrodes with CSP16mB layout on selected positions on the motor cortex of the extended 10-20 system

- Covered regions: central
- BCI experiments (motor imagery with common spatial patterns, auditory evoked potentials, vibrotactile evoked potentials)
- Simple psycho-physiological experiments focused on sensory motor cortex activation

16 electrodes with CSP16rX layout on selected positions on the motor cortex of the extended 10-20 system

- Covered regions: frontal, central
- BCI experiments (motor imagery with common spatial patterns)
- Simple psycho-physiological experiments focused on sensory motor cortex activation

Grid limitation

32 channels or less are not suitable for high density recordings for source localization. Furthermore g.Nautilus PRO is not intended for EEG recording in clinical use (clinical EEG).

Type (REF)		Number of channels
NAUTILUS 32 (REF: 5100PRO)	NAUTILUS 32 SAHARA (REF: 5200PRO)	32 channels
		
NAUTILUS 16 (REF: 5110PRO)	NAUTILUS 16 SAHARA (REF: 5210PRO)	16 channels
		

Type (REF)	Number of channels
NAUTILUS 8 (REF: 5120PRO) 	NAUTILUS 8 SAHARA (REF: 5220PRO)  8 channels
NAUTILUS CSP32 (REF: 5101PRO) 	NAUTILUS CSP32 SAHARA (REF: 5201PRO)  32 CSP channels
NAUTILUS CSP16mB (REF: 5111PRO) 	NAUTILUS CSP16mB SAHARA (REF: 5211PRO)  16 CSPmB channels
NAUTILUS CSP16rX (REF: 5112PRO) 	NAUTILUS CSP16rX SAHARA (REF: 5212PRO)  16 CSPrX channels

3 g.Nautilus PRO components

Components	WET system	DRY system
1 g.Nautilus PRO Headset - wireless biopotential amplifier with prefixed electrode strands. These strands are mounted inside an optionally available small, medium or large cap, which is not included	X	X
1 g.Nautilus PRO Base Station - stationary receiver unit	X	X
1 USB cable for Base Station (length: 2 meters)	X	X
1 Device holder shell	X	X
1 Charging station	X	X
1 USB charging cable for charging station	X	X
1 Power adapter (EU or US or UK or CN or AU)	X	X
1 Trigger cable (length: 2.90 meters)	X	X
1 Transport case	X	X
1 g.Nautilus PRO Instructions for use (PDF, downloadable)	X	X
1 Sensors and Utilities Instruction for use (PDF, downloadable)	X	X
1 Cleaning tools	X	X
1 g.Nautilus PRO Instructions for use (printed)	X	X
1 g.tec Suite 2020 (downloadable)	X	X
1 g.GAMMAgel	X	
1 g.GAMMA syringe	X	
1 set g.SAHARA gold electrodes 7 mm (8, 16 or 32)		X
1 set g.SAHARA gold electrodes 16 mm (4, 4, or 8)		X
1 set Disposable adhesive mastoid electrodes (50 pcs)		X

Accessories connected to the analog and digital interfaces must be certified according to the respective IEC standards (e.g. IEC 60950 and IEC 62368-1 for data processing equipment and IEC 60601-1 for medical equipment). Furthermore, all configurations shall comply with the system standard IEC 60601-1.

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Everybody who connects additional equipment to the signal input part or signal output part configures a medical system, and is therefore responsible that the system complies with the requirements of the system standard IEC 60601-1. If in doubt, consult the technical service department or your local representative.

4 Switches and LEDs on the Headset

g.Nautilus Headset (CSP16mB gel version shown here)



Power	Push-button for turning the device on and off
-------	---

Status	A multi-color LED indicates the status of the device
--------	--

4.1 Indications of operation mode

Status LED	Device operating mode
Short turquoise flashing (20 ms, every second)	Sender/Receiver standby state (not yet connected to Base Station)
Slow turquoise blinking (1000 ms, every 2 seconds)	Ready for transmission (connected to Base Station)
Steadily turquoise	Data transmission
Short orange flashing (20 ms, every second)	Sender/Receiver standby state (not yet connected to Base Station) capacity of accumulator < 30 %.
Slow orange blinking (1000 ms, every 2 seconds)	Ready for transmission (connected to Base Station) capacity of accumulator < 30 %.
Steadily orange	capacity of accumulator < 30 %.
Fast orange blinking	Capacity of accumulator < 10 %.

4.2 Indications of charging mode

Status LED	Device charging mode
Steadily blue	Charging of Headset
Slow blue blinking	The capacity has reached 70 % of maximum
Short blue flashing (20 ms, every second)	The Headset is fully charged

5 Markings on the top side of the g.Nautilus Headset

g.Nautilus PRO 32-channel setup (wet and dry versions)



g.Nautilus PRO 16-channel setup (wet and dry versions)



g.Nautilus PRO 8-channel setup (wet and dry versions)



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g.Nautilus PRO 32-channel setup CSP system (wet and dry versions)



g.Nautilus PRO 16-channel setup CSPmB system (wet and dry versions)



g.Nautilus PRO 16-channel setup CSPrX system (wet and dry versions)



NAUTILUS 32	g.Nautilus PRO 32-channel setup, gel version
NAUTILUS 16	g.Nautilus PRO 16-channel setup, gel version
NAUTILUS 8	g.Nautilus PRO 8-channel setup, gel version
NAUTILUS CSP32	g.Nautilus PRO 32-channel setup CSP system, gel version
NAUTILUS CSP16mB	g.Nautilus PRO 16-channel setup CSPmB system, gel version
NAUTILUS CSP16rX	g.Nautilus PRO 16-channel setup CSPrX system, gel version
NAUTILUS 32 SAHARA	g.Nautilus PRO 32-channel setup, dry version
NAUTILUS 16 SAHARA	g.Nautilus PRO 16-channel setup, dry version
NAUTILUS 8 SAHARA	g.Nautilus PRO 8-channel setup, dry version
NAUTILUS CSP32 SAHARA	g.Nautilus PRO 32-channel setup CSP system, dry version
NAUTILUS CSP16mB SAHARA	g.Nautilus PRO 16-channel setup CSPmB system, dry version
NAUTILUS-CSP16rX SAHARA	g.Nautilus PRO 16-channel setup CSPrX system, dry version
	CE mark
	Do not dispose g.Nautilus with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment.
	Applied part BF
	Safety class II
	Avoid electrostatic discharge
	Follow instructions for use



Standby for a part of the device



g.tec medical engineering GmbH
Sierningstrasse 14
4521 Schiedlberg, Austria
www.gtec.at

Manufacturer address

IP44

IP 44: Protected against objects >1 mm.
Water splashing against the enclosure from any direction shall have no harmful effect.

S1

S1: permanent operation

FCC ID: XVV-MEGA22M00

FCC-ID of included transmission module

Contains IC: 8720A-22M00

IC-ID of included transmission module

Rx only

Prescription device in the USA

SN NP-2019.05.01

Serial number in the format:
NP-YearOfProduction.Month.Number
Must be the same as on Base Station.

REF 5120PRO

Catalog Reference

+EBCI5120PRO0/\$+NP-2019.05.019

UDI Human readable



UDI code in automatic identification and data capture (AIDC) format

6 g.Nautilus PRO Base Station, LED and USB cable

The Base Station and the corresponding USB-cable, together with a connected PC (certified according to the respective IEC standards (e.g. IEC 60950 or IEC 62368-1 for data processing equipment, respectively)) and driver, allow communication with the g.Nautilus Headset.

g.Nautilus Base Station
(for the CSP16mB gel version shown here)

USB cable



STATUS:

A turquoise LED indicates the status of the device

Type

Number of channels

NAUTILUS 32

NAUTILUS 32 SAHARA



32 channels

NAUTILUS 16

NAUTILUS 16 SAHARA



16 channels

Type	Number of channels	
NAUTILUS 8	NAUTILUS 8 SAHARA	8 channels
		
NAUTILUS CSP32	NAUTILUS CSP32 SAHARA	32 CSP channels
		
NAUTILUS CSP16mB	NAUTILUS CSP16mB SAHARA	16 CSPmB channels
		
NAUTILUS CSP16rX	NAUTILUS CSP16rX SAHARA	16 CSPrX channels
		

6.1 Indications of operation mode

Status LED	Device operating mode
Short turquoise flashing (20 ms, every second)	Sender/Receiver standby state (not yet connected to Headset)
Slow turquoise blinking (1000 ms, every 2 seconds)	Ready for transmission (connected to Headset)
Steadily turquoise	Data transmission

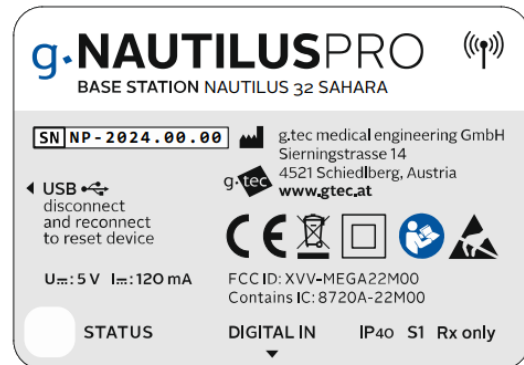
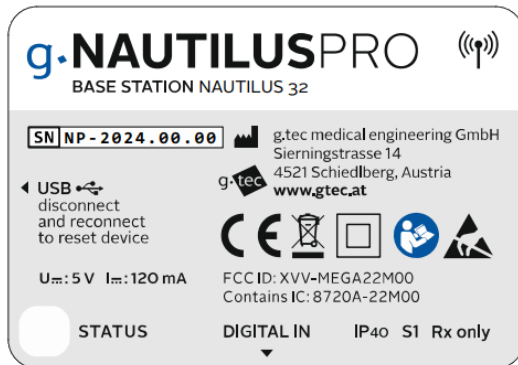
Part of the g.Nautilus medical equipment is the USB cable. It is a 4 pin cable used to connect the Base Station to a PC. One side has a standard USB connector the other side a Mini-USB connector.

6.2 LABELS ON THE CABLE

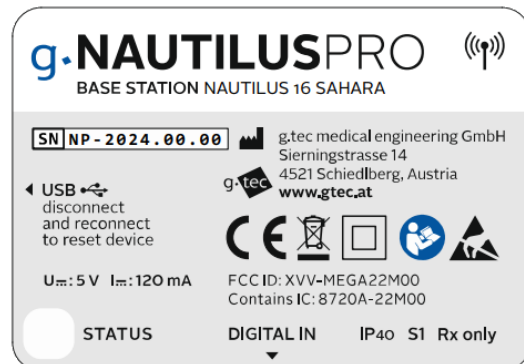
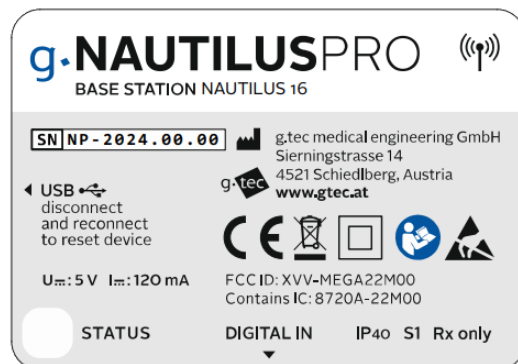
USB - PC	USB connector to PC
g.Nautilus PRO Base Station	USB connector to Base Station

7 Markings on the top side of the g.Nautilus Base Station

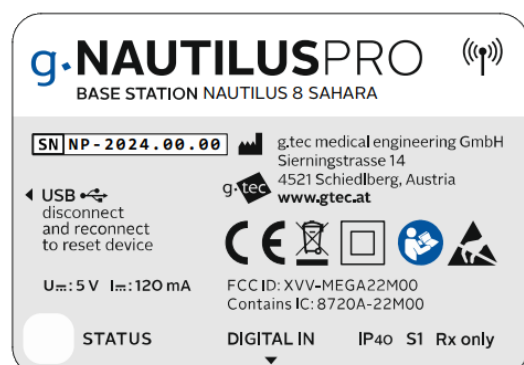
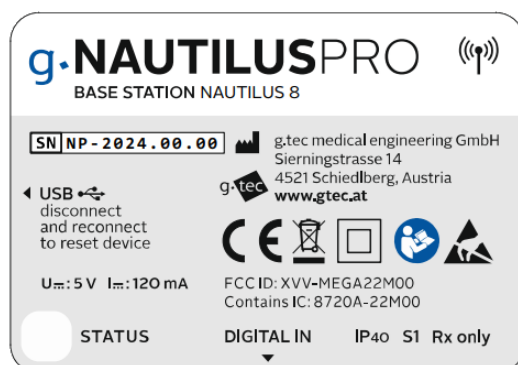
g.Nautilus PRO 32-channel setup (wet and dry versions)



g.Nautilus PRO 16-channel setup (wet and dry versions)

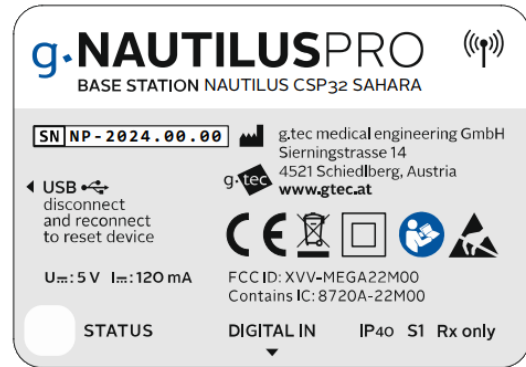
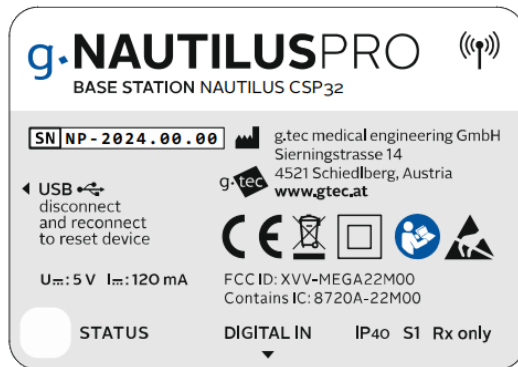


g.Nautilus PRO 8-channel setup (wet and dry versions)

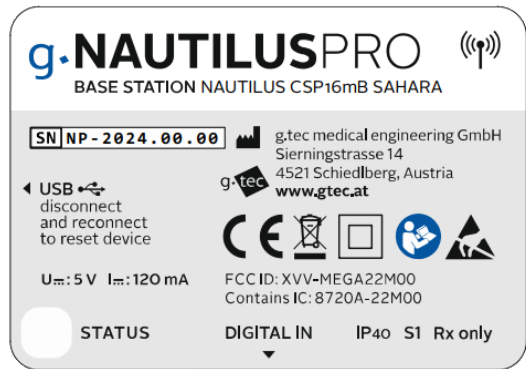
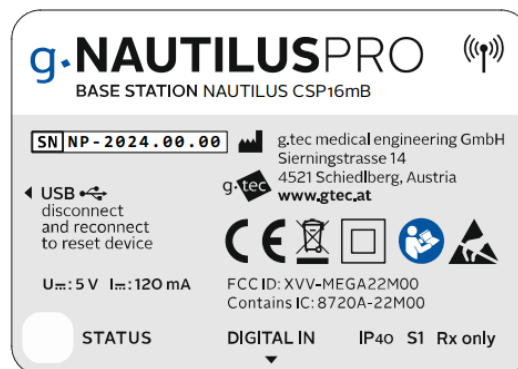


g.NAUTILUSPRO

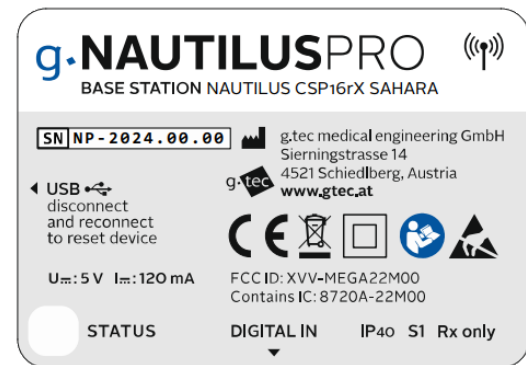
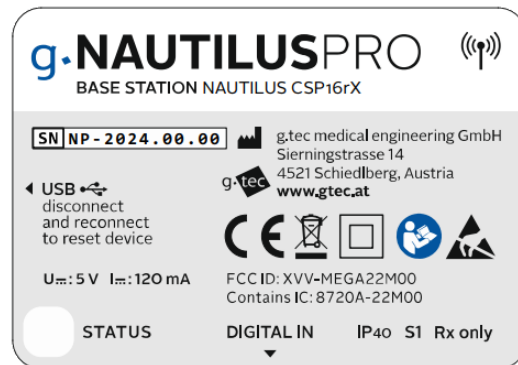
g.Nautilus PRO 32-channel setup CSP system (wet and dry versions)



g.Nautilus PRO 16-channel setup CSPmB system (wet and dry versions)



g.Nautilus PRO 16-channel setup CSPrX system (wet and dry versions)



NAUTILUS 32	g.Nautilus PRO 32-channel setup, gel version
NAUTILUS 16	g.Nautilus PRO 16-channel setup, gel version
NAUTILUS 8	g.Nautilus PRO 8-channel setup, gel version
NAUTILUS CSP32	g.Nautilus PRO 32-channel setup CSP system, gel version
NAUTILUS CSP16mB	g.Nautilus PRO 16-channel setup CSPmB system, gel version
NAUTILUS CSP16rX	g.Nautilus PRO 16-channel setup CSPrX system, gel version
NAUTILUS 32 SAHARA	g.Nautilus PRO 32-channel setup, dry version
NAUTILUS 16 SAHARA	g.Nautilus PRO 16-channel setup, dry version
NAUTILUS 8 SAHARA	g.Nautilus PRO 8-channel setup, dry version
NAUTILUS CSP32 SAHARA	g.Nautilus PRO 32-channel setup CSP system, dry version
NAUTILUS CSP16mB SAHARA	g.Nautilus PRO 16-channel setup CSPmB system, dry version
NAUTILUS CSP16rX SAHARA	g.Nautilus PRO 16-channel setup CSPrX system, dry version
	CE mark
	Do not dispose g.Nautilus with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment.
	Avoid electrostatic discharge
	Follow instruction for use
	Safety class II
 g.tec medical engineering GmbH Sierningstrasse 14 4521 Schiedlberg, Austria www.gtec.at	Manufacturer address
IP40	IP40: Protected against objects >1 mm. Not protected against water.
S1	S1: permanent operation
SN NP-2019.05.01	Serial number in the format: NP-YearOfProduction.Month.Number Must be the same as on Headset.
FCC ID: XVV-MEGA22M00	FCC-ID of included transmission module
Contains IC: 8720A-22M00	IC-ID of included transmission module
DIGITAL IN ▼	D-SUB connector to connect digital input signals

 USB disconnect and reconnect to reset device	Mini-USB connector to connect computer
	Radio transceiver inside
Rx only	Prescription device in the USA
U_~: 5 V	Rated voltage
I_~: 120 mA	Rated current

8 Charging device

The g.Nautilus medical equipment includes a wireless charging pad and a power supply that is connected via a USB cable. The power supply delivers 5 V DC to the charging pad, which wirelessly charges the g.Nautilus. The USB cable has one USB-A connector and one micro USB port.

8.1 Charging pad and USB cable

Charging device with cable



8.1.1 Labels on the charging pad

PowerWave Pad	Type
Model	A2503
Input: 5 V $\Rightarrow$ 2 A / 9 V $\Rightarrow$ 2 A Input voltage and current	
Output: 10 W Max.	Output power
FCC ID	2AOKB-A2503D
IC:	23451-A2503D
Anker Innovations Limited	Manufacturer
Made in China	Origin
SN:	Serial number of the charging pad

	Do not dispose with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment
	CE mark
	Regulatory Compliance Mark (RCM)
	UKCA mark
	interface standard developed by the Wireless Power Consortium

8.1.2 Labels on the USB cable

Wall adapter	USB connector to power supply
Charging Pad	Micro USB connector to charging pad

8.2 Power Supply for charging pad

Power supply with blades for different countries



8.2.1 Labels on the power supply

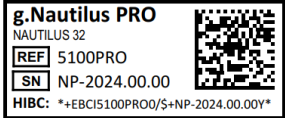
CUI Inc.	Manufacturer
Model: SMI10-5-V-I38	Model code power supply without blades
Input: 100-240V~, 50/60Hz, 0.3 A	Input voltage, frequency and current
Output: +5 V $\overline{\text{=}}$ 2A, 10.0 W	Output voltage, current and power
Blades: SMI-US-3L SMI-CN-3L SMI-UK-3L SMI-EU-3L SMI-AU-3L	Region: North America and Japan China UK Europe Australia
Made in China	Origin
	Regulatory Compliance Mark (RCM)

	CE mark
	UKCA mark
	FCC mark
	C-UL-US mark
	China Compulsory Certificate mark
	Product Safety Electrical Appliance & Material mark
	Do not dispose with domestic waste. Dispose it via the separate collection system for electrical and electronic equipment
	Safety class II
	Energy efficiency level 6
	For indoor use only

The corresponding blade for the specific region is pre-attached in the delivery content.

9 Packaging

The figures below show the packaging of the device and the labelling of the box.

g.Nautilus Case	Case Label	UDI Label
		

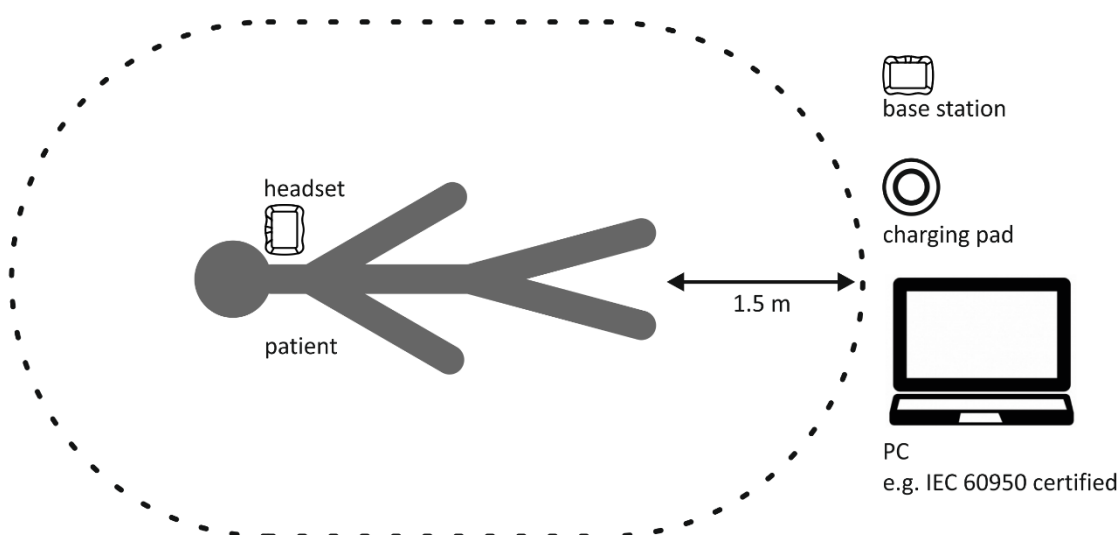
10 Safe operation of g.Nautilus

10.1 Set up and measurement with the device

- Use on healthy, intact skin only.
- Do not use on or near open wounds, bruised or weakened skin, whether due to injury and/or other medical conditions of the patient.
- Do not use on patients with a history of skin allergies or sensitivity to cosmetics and lotions.
- If rash, redness, itching, swelling, or abnormality appears on skin, remove the cap and wash any residual gel off immediately.
- Instruct patients to communicate any persistent redness, soreness or swelling at the electrode sites.
- It is not recommended that the cap be left in place for longer than eight hours for any single recording session.
- Avoid eye contact with gel as it may cause mild transient irritation. Wash eyes with appropriate 0.9 % saline solution, commercial eye wash solution or water to flush out residual particles. Avoid rubbing eyes.
- Electro-Gel is essentially non-toxic, however ingestion could result in nausea, vomiting and/ or diarrhea. Drink 1 to 2 glasses of water if ingested.

10.2 Patient environment

- The Base Station MUST be placed outside the patient environment.
- The PC or notebook and UPS⁴ (certified according to the respective IEC standards (e.g. IEC 60950 or IEC 62368-1 for data processing equipment)) MUST be placed outside the patient environment.
- The charging pad MUST be placed outside the patient environment, even the device is not charging.



⁴ uninterruptible power supply

10.3 Avoiding electrostatic discharge

Electrostatic discharge (ESD) events can harm electronic components inside your device. Under certain conditions, electric charge may build up on your body or an object, such as a peripheral, and then discharge into another object, such as your device. To prevent ESD damage, you should discharge static electricity from your body before you interact with any of your devices.

You can protect against ESD and discharge static electricity from your body by touching a metal grounded object.

Please perform the following steps to record data with g.Nautilus:

Step 1: Place the g.Nautilus Headset on the head of the subject to position the electrodes according to the international 10/20 electrode system⁵.

In order to properly mount the electrode cap, measure the distance between the Nasion and Inion and the distance between the left and right preauricular points of the subject. The position at the middle of these two locations is the vertex position, Cz. Put on the electrode cap and align this measured position with the position Cz of the electrode cap. Now, the electrode cap is in the correct position.

Fix the cap with the chin strap to keep the cap in place.

Step 2: Mount EEG electrodes

10.4 g.Nautilus PRO: Gel electrodes

Fill all electrodes with conductive gel. You must coat at least 1 cm² of skin under the electrode with gel and the electrode must be filled completely. The gel is best applied with a syringe. Insert the tip of the syringe into the ready-mounted electrode and make sure to reach the skin. Press the syringe and move it around a bit with the tip on the skin, with slight pressure to make sure that gel is distributed on the skin. Then carefully fill the electrode with gel. It is essential that gel completely fills the gap between the skin and the electrode surface! However, using too much gel may cause short circuits between electrodes.

Depending on the used software you may have access to the built-in impedance measurement (works only for gel-based electrodes!).

The electrode impedance should be below 50 kOhm (yellow channel indication software) or best below 30 kOhm (green channel indication software). Choose appropriate settings in your software and check the signal quality of all channels.

The Z-channel electrode (which is shown/highlighted in the Appendix and is depending on the channel setup) is used for the determination of the electrode impedances and should also be mounted very carefully. Impedance values of the Z-channel, the reference electrode and ground electrode are not determined by the impedance check. During the impedance measurement it is important that the Z- and GND electrode have a good contact with the skin.

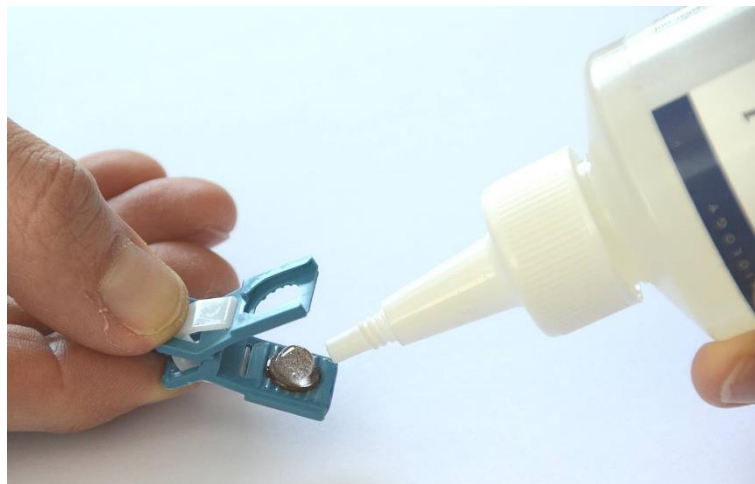
⁵Webster, J.G., (Ed.): Medical Instrumentation: Application and Design, p.194-216. Houghton Mifflin: Boston 1992.

Inject electrode gel into the active, gel electrodes.



The ear-clip electrode is attached with gel as well. Just put a small amount of gel onto the electrode surface (Ag/AgCl disk) and put the earclip on the earlobe, with the electrode outside.

Reference electrode



If a signal shows high noise, drift or other disturbance, check the proper filling of the electrode ring and assure that the gel touches the skin. Note that the ground electrode (yellow) is a passive electrode, and therefore it is the only electrode that needs to be mounted with special care. Very high impedance at the ground electrode may reduce the signal quality. One can use abrasive gel to prepare the skin at the ground electrode if necessary.

10.5 g.Nautilus PRO: Dry electrodes

The dry electrodes (g.tec medical engineering GmbH, Austria) have 8 golden pins with 7 mm (standard configuration) or 19 mm length.

Dry electrode



The electrodes can be clipped into the g.Nautilus PRO dry active electrode system.

Electrode cap with inserted dry electrodes



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For short and normal hair, use the standard version. For long and dense hair, use the long-pin version. Use a cap size (S, M or L) that provides gentle pressure to all electrodes on the skin without causing pain to the subject. Rotate each electrode back and forth a few times at an angle of 10° - 20° to remove hair between the electrodes and the skin, and to ensure a proper galvanic contact between the electrode pins and the skin.

Mount the blue reference electrode on the right mastoid with disposable Ag/AgCl electrodes. Mount the ground electrode on the left mastoid, also with disposable electrodes.



NOTE

After 4-5 minutes at most, all signals should be visible and show a stable baseline. The impedance measurement function does not work for dry electrodes. In case of high drifts or noisy signals on a certain channel, the corresponding electrode needs to be checked for good contact with the skin. Place gentle pressure on the electrode and move it a little bit on the skin, then wait again for 1 minute and check the signal again.



NOTE

Dry electrodes can produce a high offset voltage. To be sure not to saturate the amplifier, adjust the input voltage range to ± 375 mV or to ± 562.5 mV for all channels. Saturating the input would lead to flat-line signals.



NOTE

Special requirements for high quality EEG recording with dry electrodes:

EEG recordings in general are quite sensitive to a number of disturbing influences such as movements of the subject, the cables and the electrodes, or coupling of electromagnetic and electrostatic fields near the recording environment. For dry electrodes especially, movements of the electrodes and electrostatic charges can cause significant artefacts in the signals. Consider the following actions to avoid artefacts:

10.6 Avoiding electrode and cable movements

- ensure the subject is in a comfortable and relaxed position during recording
- instruct the subject to stay in a relaxed position and to avoid movements
- place electrode wires to avoid tension and direct contact with moving parts of the body
- if the subject needs to speak during the recordings, use the chest belt rather than the chin strap to fixate the cap

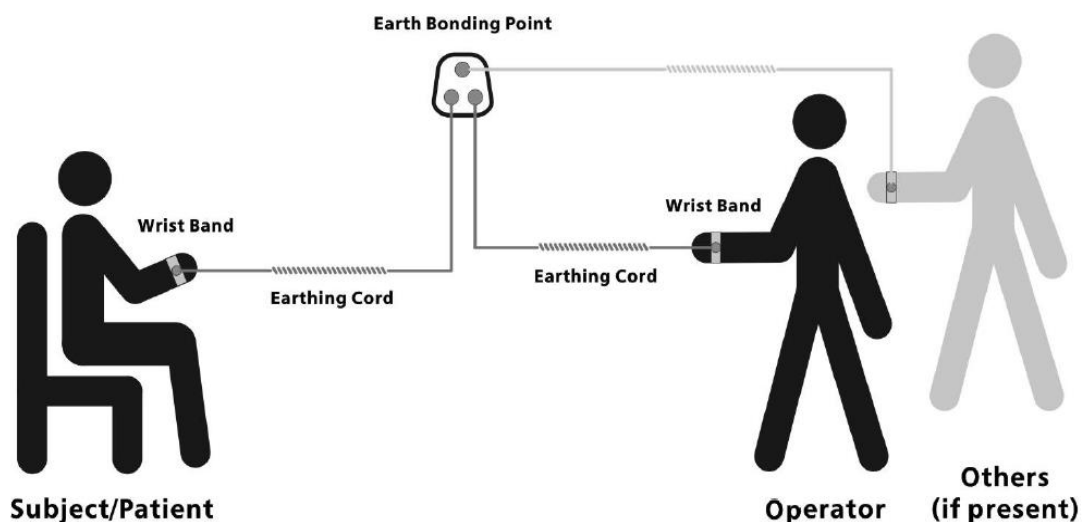
10.7 Avoiding electrostatic charges

- choose an antistatic environment/lab for high quality recordings
- use a room with antistatic carpet or floor covering or wooden/stone floor instead of normal carpets or artificial parquet floors
- furniture made of wood, metal and leather is preferable to any plastic materials and synthetic textiles
- cotton clothing is preferred to wool and synthetics
- avoid shoes with synthetic soles

10.8 Grounding

- grounding of any additional equipment (using a potential equalization conductor) and metal-made furniture to a central earth/ground contact (lab ground) is recommended
- the subject as well as the operator(s) may be grounded to avoid electrostatic charges (e.g. using an antistatic wrist band connected to the lab ground)

Example of recommended connection, including antistatic wrist bands from the subject and the operator(s) to one common earth bonding point.



Step 3: Press the power button on the Headset for at least 3 seconds to switch on the device. During the button press, the STATUS LED must be solid turquoise. After switching on the device, the STATUS LED must flash quickly in turquoise. The Headset is not yet connected to the Base Station.

Contact the manufacturer if the LED is not on, even after the accumulator has been fully charged.

Step 4: Connect the Base Station to a free USB 2.0 port on your computer using the USB cable. After a short time period (around 3 seconds), the STATUS LED of the Headset should blink slowly. The Headset has automatically connected to the Base Station.

The STATUS LED of the Headset should slowly blink in turquoise. If the STATUS LED is orange, the Headset must be charged.

The STATUS LED indicates that g.Nautilus PRO is operating correctly. If the STATUS LED is not on, please check the charge of the accumulator by pressing the power button, or contact the manufacturer.



ATTENTION

- The Base Station **MUST** be placed outside the patient environment.
- The PC or notebook and UPS (certified according to the respective IEC standards (e.g. IEC 60950 or IEC 62368-1 for data processing equipment, respectively)) **MUST** be placed outside the patient environment.

Step 5: Connect digital inputs to the Base Station if synchronization of the EEG data with external devices is required.

Step 6: Check the impedance of the electrodes to ensure a proper contact between the electrodes and the skin. An internally generated square wave (+15 mV) is supplied to the ground electrode to measure the impedance at 10 Hz.

Step 7: Adjust the bandpass filters and Notch filters for the recording channels, specify any bipolar or common average reference (CAR) derivations, and select the sampling frequency, noise reduction and digital inputs. For gel electrodes, use a sensitivity setting of 187.5 mV (highest resolution). For dry electrodes, due to polarization effects causing DC offsets, use 562.5 mV.

Step 8: Perform the EEG measurement with the g.Nautilus according to your experimental procedure. In measuring mode, the STATUS LED indicates the data transfer with a permanent turquoise color. All input channels (referred to the REF channel) are amplified, sampled at 250 Hz or 500 Hz, and converted with a 24 bit analog to digital converter. The digitized data are sent via the wireless data link to the Base Station and via USB to the computer.

10.9 Switch off and store the device

To switch off and store g.Nautilus Headset correctly, please perform the following steps:

Step 1: Switch off the Headset by pressing the push-button for at least 3 seconds. The STATUS LED will be on steadily during the button press, and will turn off after about 3 seconds.

Step 2: Carefully remove the Headset with the cap and the electrodes from the subject's head and clean the electrodes immediately. The gel electrodes can be cleaned under warm water. The dry electrodes can be cleaned with cleaning alcohol (do not use water for the dry electrodes).

Step 3: Disconnect the USB-cable of the Base Station and any trigger cables.

Step 4: Remove the device from the device holder shell (see Figure 1), then put the device and all the accessory into the corresponding slot in the case (see Figure 2). Pay attention that the device and the cap are not wet when they are stored. Also the wires mustn't be buckled and wedged-in when the case is closed.

Figure 1: remove the device from the holder shell



Figure 2: Device and accessory in the case



10.10 Charging of g.Nautilus

The g.Nautilus Headset is charged via an inductive charging technology according to the Qi standard of the wireless power consortium⁶. Position the Headset (switched on or off) on a Qi compatible charging pad. Connect the charging pad with the delivered power supply and USB cable to a mains receptacle.

⁶<http://www.wirelesspowerconsortium.com/>

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Charging starts automatically and is indicated via the permanently blue glowing STATUS LED. A blue blinking STATUS LED indicates that the capacity has reached 70 % of its maximum. If the STATUS LED is flashing very quickly, the accumulator is fully charged. The accumulator cannot be overcharged, so the Headset can stay on the charging pad without being damaged.



ATTENTION

- The charging pad can detect whether a device is close that can be charged. Therefore, it is important to put the g.Nautilus with label facing up onto the charging pad.
- The charging pad **MUST** be placed outside the patient environment, even the device is not charging

10.11 Details for the correct maintenance of the accumulator

The g.Nautilus Headset has a built-in Lithium-Ion accumulator. When the device is delivered, the accumulator is partly charged. Before first use, fully charge the device until the STATUS LED starts flashing. If the device is not used for more than 2 weeks, take care that the accumulator is always charged between 50 and 90 %. This will maximize the accumulator's lifetime. So, before longer storage, the device should be charged until the STATUS LED blinks slowly. If the device is stored for more than 8 weeks, a risk of deep discharge of the accumulator exists, which could lead to damage! For this reason, the device must be charged every 4 weeks until the STATUS LED blinks slowly.

If the device is not switched off after use, it will run until the voltage of the accumulator drops to a certain value and will be switched off automatically. In this case, the device should be charged as soon as possible to prevent deeper discharge and damage to the accumulator.

The longest possible lifetime of a Lithium-Ion accumulator is achieved if it is charged to 100 % and then discharged to less than 20-30 % as rarely as possible. If feasible under the operational conditions, the accumulator should be charged if the STATUS LED color changes from turquoise to orange, and charging should be stopped when the STATUS LED starts to blink slowly. If the maximum operation time is required, continue charging till the STATUS LED flashes quickly.

Charging of g.Nautilus Headset via a Qi-compatible charging pad.



10.12 Cleaning and Disinfection of Headset, cap and electrodes



NOTES

- The electrode caps do not have an expiration date. Elasticity of the material dictates the life of the cap. When a cap becomes stretched, it loses its elasticity. It must then be discarded and replaced, as the electrode placements will be inaccurate and artifacts may occur more often.
- Do not submerge the device in water, and do not expose it to powerful streams of water such as some cleaning jets. Always remove the Headset device from the holder shell when washing the cap to protect it against liquids. The holder shell may remain on the cap.
- Please note that devices that blow warm or hot air, such as devices to dry hands or hair, must not be used to dry electrode caps, because it weakens the elastic material; the cap lifetime will be greatly shortened. Hanging the cap in front of a fan will accelerate drying.
- The g.Nautilus PRO is a noncritical patient contact device.
- Take care that the electrode leads are not damaged during cleaning!
- Do not use excessive force while cleaning.
- The device itself must not come in contact with germicide.
- It is the user's responsibility to qualify any deviations from the recommended method of processing. Users should note appropriate disclaimers if there are deviations. g.tec medical engineering GmbH assumes no responsibility for damage to its products caused by the use of products or processes not recommended in this policy.
- If questions arise, or more information is needed, please call g.tec medical engineering GmbH for assistance:
 - phone: +43 7251 22240-0
 - fax: +43 7251 22240-39
 - mail: office@gtec.at

11 Initial instructions on how to make the device patient ready

Clean the g.Nautilus PRO before using it for the first time, as described below for gel electrodes or dry electrodes. Check before use that no electrode leads or housing are damaged!

11.1 Removing gel

- When the EEG recording is finished, remove the cap and ear electrodes.
- Wipe the patient's forehead and ears with a tissue or cloth towel, and lightly wipe the hair. Brushing or combing the hair will remove all visual evidence of the gel. The patient's next shampoo will easily and completely eliminate any remaining gel.

11.2 Containment and Transport

Prior to disinfection (when required), transport the cap and equipment safely (such as in a closed box) and wear appropriate gloves for safety (see Figure 4).



ATTENTION

Ag/AgCl electrodes and dry electrodes do not have an expiration date, but careful treatment prolongs life time! When an electrode becomes noisier and artifacts occur more often, it must be replaced by the manufacturer only.

11.3 Cleaning g.Nautilus PRO: Gel electrodes and cap

After usage, the Headset must be cleaned immediately to prevent the desiccation of gel. All steps must be performed thoroughly.

EQUIPMENT: Detergent: multistage enzymatic cleaner (e.g. Sekusept® MultiEnzyme P #3046670 from ECOLAB (www.ecolab.com) or your facility may have a similar product), lukewarm water (20 °C to 25 °C) of at least drinking water quality, soft brush (e.g. Interlock #09068, #09084), detergent tank.

- The cleaner should be prepared in accordance to the manufacturer's instructions with a concentration of 2 %.
- Detach the device carefully from the device holder shell, but leave the electrodes in the cap. The holder shell may remain on the cap for cleaning and disinfection.
- The device is protected against water splashes, but must not be submerged in water or any fluid!
- Immerse the submersible part of the device into the detergent (see Figure 3).
- Soak for at least 30 minutes.
- Clean all electrodes inside and outside with the brush until all gel is removed (see Figure 4).

Figure 3: bowl with cap and electrodes inside



ATTENTION

Do not submerge the device!

Figure 4: cleaning of the electrodes with the toothbrush



- Remove the device from the detergent.
- Immerse the submersible part of the device into a tank with tap water and rinse the cap and the electrodes thoroughly.
- Visually inspect the cap and each electrode and repeat the preceding steps if necessary.

11.4 Disinfection of g.Nautilus PRO: Gel electrodes and cap

Equipment: Low Level disinfectant based on Glucoprotamin (e.g. Sekusept® PLUS #3011100 from ECOLAB (www.ecolab.com) or your facility may have a similar product), lukewarm water (20 °C to 25 °C) of at least drinking water quality, disinfectant tank.

- The disinfectant should be prepared in accordance to the manufacturer's instructions.
- Immerse the submergible part of the device into the detergent (see Figure 3).
- Continue soaking as long as recommended in the manufacturer's instructions.
- Remove the cap and electrodes from the disinfectant bath.
- Rinse the device under tap water for at least 30 seconds (at least 1 liter).
- Put one towel inside the cap and one towel outside to remove the water.
- Put the cap in an appropriate place for drying (see Figure 5). The cap will dry in about one hour.
- Avoid stretching the cap's fabric during the process of drying.
- Take care that all components are completely dry before storing or packing g.Nautilus PRO.

Figure 5: one example of a way to dry the wet electrode system



11.5 Cleaning g.Nautilus PRO: Dry electrodes and cap

All steps must be performed thoroughly.

Equipment: Detergent: multistage enzymatic cleaner (e.g. Sekusept® MultiEnzyme P #3046670 from ECOLAB (www.ecolab.com) or your facility may have a similar product), lukewarm water (20 °C to 25 °C) of at least drinking water quality, detergent tank.

- The cleaner should be prepared in accordance to the manufacturer's instructions.
- Detach the dry electrodes and the device with the wires from the cap (see Figure 6). The device holder shell may remain on the cap.
- The device is protected against water splashes, but must not be submerged in water or any other fluid!
- Immerse the submersible part of the cap into the detergent (see Figure 7).
- Soak for at least 30 minutes.

Figure 6: assembly and disassembly of the wires from the dry electrodes



ATTENTION

Do not submerge the device!

- Remove the cap from the detergent.
- Immerse the submersible part of the cap into a tank with tap water and rinse the cap thoroughly.
- Visually inspect the cap and each electrode and repeat the preceding steps if necessary.

Figure 7: Submergible part

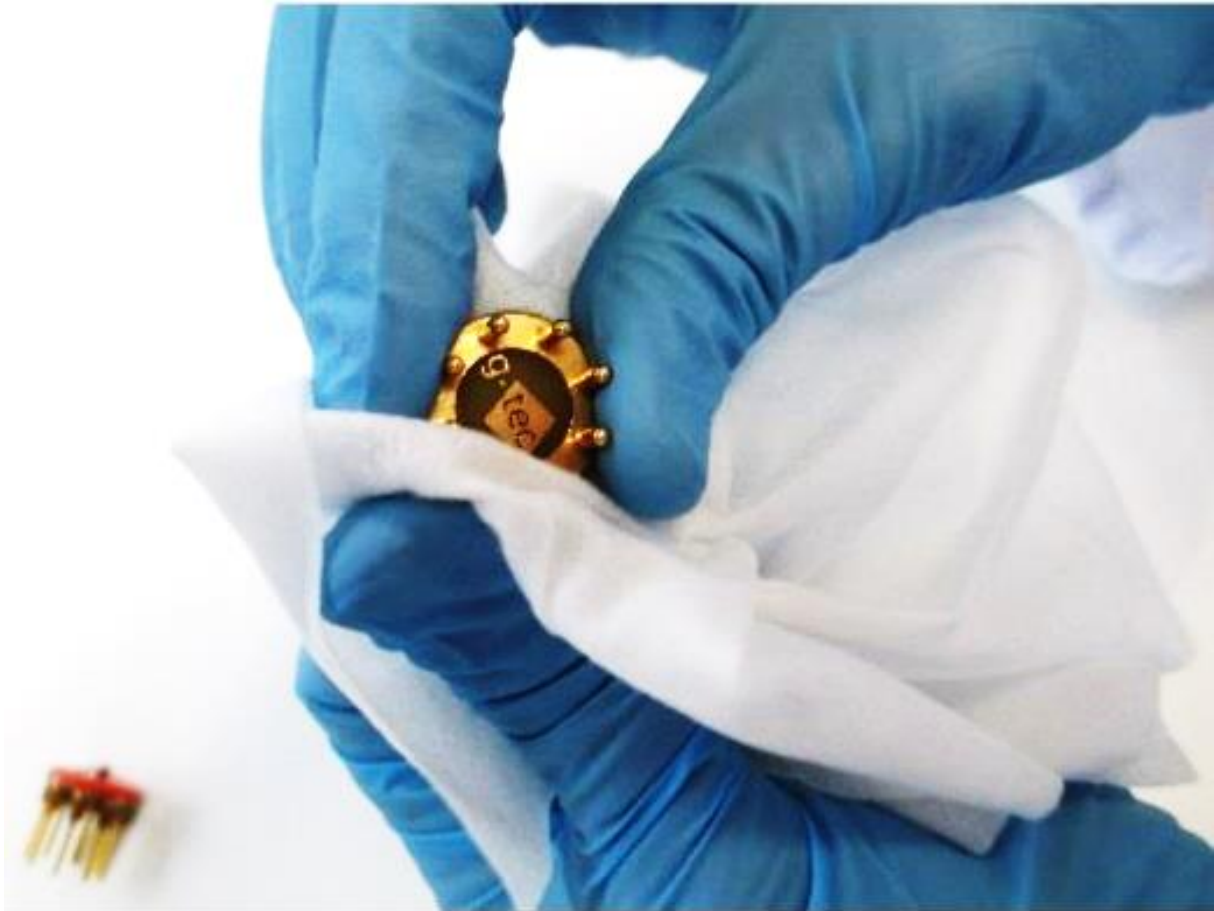


11.6 Disinfection g.Nautilus PRO: Dry electrodes and cap

Equipment: cleaning and disinfection wipes (e.g. Mikrozid® sensitive wipes #70000807, Mikrozid® AF wipes #109203 from Schülke & Mayr GmbH (www.schuelke.com), or your facility may have a similar product).

- Clean the pins of the dry electrodes with the cleaning wipe (see Figure 8).
- Be sure to wipe all surfaces thoroughly.
- Take a disinfection wipe and repeat step 1 and 2.
- Don't use water.
- When the cap is dry, reassemble the device and the electrodes on the cap.
- Take care that all components are completely dry before storing or packing g.Nautilus PRO.

Figure 8: cleaning of the pins



Equipment: Low Level disinfectant based on Glucoprotamin (e.g. Sekusept® PLUS #3011100 from ECOLAB (www.ecolab.com) or your facility may have a similar product), lukewarm water (20 °C to 25 °C) of at least drinking water quality, disinfectant tank.

- The disinfectant should be prepared in accordance to the manufacturer's instructions.
- Immerse the submersible part of the device into the detergent (see Figure 7).
- Continue soaking as long as recommended in the manufacturer's instructions.
- Remove the device from the disinfectant bath.
- Rinse the device under tap water for at least 30 seconds (at least 1 liter).
- Put one towel inside the cap and one towel outside to remove the water.
- Put the cap in an appropriate place for drying (see Figure 5). The cap will dry in about one hour.
- Avoid stretching the cap's fabric during the drying process.
- Take care that all components are completely dry before storing or packing g.Nautilus PRO.

11.7 Cleaning and Disinfection of enclosure parts of Headset, Base Station and charging pad

Equipment: cleaning and disinfection wipes (e.g. Mikroqid® sensitive wipes #70000807, Mikroqid® AF wipes #109203 from Schülke & Mayr GmbH (www.schuelke.com), or your facility may have a similar product).

- Clean the device enclosure with the cleaning wipes.

- Be sure to wipe all surfaces thoroughly and keep them wet for the whole exposure time.
- Take a disinfection wipe and repeat steps 1 and 2.
- Ensure that all components are completely dry before storing or packing.



ATTENTION

- During cleaning and disinfection, the Base Station must be disconnected from the PC.
- During cleaning and disinfection, the charging pad must be disconnected from the main voltage supply.
- Do not spill fluids on the Headset, Base Station, or charging pad.

11.8 How to change or replace electrode caps of different sizes (S/M/L) on a g.Nautilus PRO system

Damaged caps can be replaced on a g.Nautilus PRO system, but should be done by the manufacturer or by corresponding qualified personnel only. The system is not designed to change between different cap sizes frequently! When changing or replacing a cap, make sure to insert the electrodes at the correct positions as shown in the appendix of this document.



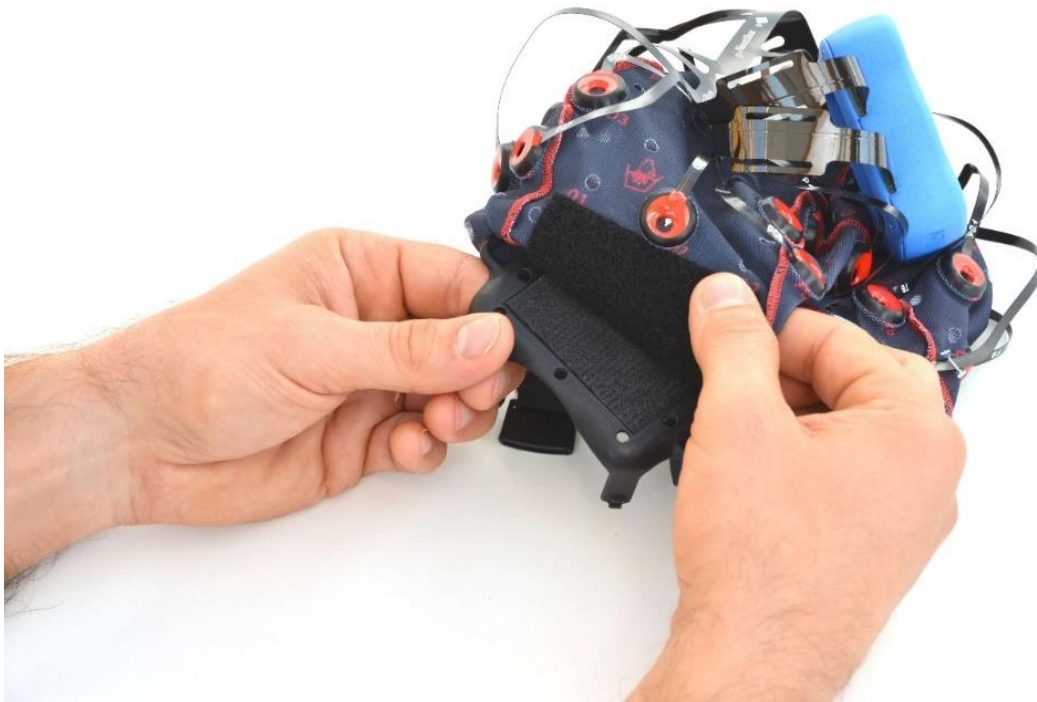
CAUTION

Make sure not to kink, stress or damage the flexible PCB wires during this process!

Step 1: Detach all electrodes from the cap. For g.SAHARA electrodes, just remove the clip connector from the electrode to release it. For g.LADYbird electrodes, stretch the fabric of the cap and pull out the electrode towards the wire side (outer side of the cap). Take the electrode between the thumb (outer side of the cap) and index finger (inner side of the cap) of one hand and stretch the fabric with the other hand. Then tilt the electrode a bit to get the fabric out of the notch on one side. Lift the electrode to remove it completely from the cap.

Step 2: Detach the Headset from the device holder shell first (see Figure 1). Then remove the holder shell from the cap by carefully releasing the hook-and-loop fastener connection between the enclosure and the cap (see Figure 9).

Figure 9: Detach the holder shell from the old cap



Step 3: Attach the device holder shell to the new cap by placing it on the hook-and-loop fastener pad. Press the hook-and-loop fastener of the new cap against the device holder shell.

Step 4: Attach the electrodes to the new cap. For g.SAHARA electrodes just place the electrode in the correct position with the contact pin through the hole in the cap and press the clip connector on the pin. For g.LADYbird electrodes place the electrode at the correct position and hold it again with two fingers, one outside the cap on the electrode and one inside the cap under the electrode. Insert the fabric of the cap into the notch on the electrode at one side. Then stretch the fabric to the opposite side and allow the fabric to slide into the notch.

Step 5: Put the cap on a polystyrene head and adjust the directions of the cable outlets according to the geometry of the flexible PCB for all electrodes.

12 General notes

12.1 Classification

Safety Class	II
Type of applied part	BF
Protection against mechanical distortion and liquids (Headset)	IP44
Protection against mechanical distortion and liquids (Base Station)	IP40
Operation mode	S1 (Permanent operation)
Conformity class	Ila

12.2 Transportation and storage conditions

The device can be stored and transported at temperatures between -20° to $+45^{\circ}$ Celsius. The relative humidity must be between 25 % and 80 %. If there is any condensed water, wait until it disappears before use (wait at least 1 h in a heated room).

12.3 Location details

Do not use the device near a heating system or directly in the sun. During operation, the outside temperature should be between $+5^{\circ}$ Celsius and $+35^{\circ}$ Celsius and the air pressure between 700 and 1060 hPa.

12.4 Waste disposal details

Bring the device to a recycling center or sent it back to the manufacturer.

12.5 Warranty

One-year warranty

13 Declaration of conformity

The declaration of conformity is available on request.

14 Technical specifications

Model	g.Nautilus PRO
Types	NAUTILUS 32 NAUTILUS 32 SAHARA NAUTILUS 16 NAUTILUS 16 SAHARA NAUTILUS 8 NAUTILUS 8 SAHARA NAUTILUS CSP32 NAUTILUS CSP32 SAHARA NAUTILUS CSP16mB NAUTILUS CSP16mB SAHARA NAUTILUS CSP16rX NAUTILUS CSP16rX SAHARA
Accumulator	PA-L36.K06.R001, 1300 mAh, IEC 62133(ed.2)
Rated power consumption	0.5 W
Rated DC voltage	3.7 V
Rated current of fuse	Little fuse 0467001.NR (1.0 A)
Rated voltage of fuse	32 V
Produced	see serial number of g.Nautilus Headset
Manufacturer	g.tec medical engineering GmbH Sierningstrasse 14 4521 Schiedlberg Austria http://www.gtec.at

14.1 Amplifier Settings

Channels 1 to 32 and REF channel

Sensitivity	$\pm 187.5 \text{ mV to } \pm 2.25 \text{ V}$
Highpass	0 Hz
Lowpass	10.23 kHz
Input impedance:	$> 1 \text{ G}\Omega$

14.2 Acceleration sensor

x-, y- and z-axis	$\pm 6 \text{ g}$ (bias of 0 – 0.4 g)
-------------------	---------------------------------------

14.3 Analog-Digital-Converter (ADC)

Resolution	24 Bit
Sampling frequency	250 / 500 Hz
Number of ADCs	32

14.4 Analog signal processing time and digital inputs

Sampling frequency (Hz)	ASD AI/DI ^{*)} (Samples)
250	2 to 3 + 0 to 11
500	2 to 3 + 0 to 11

*) Analog Signal line Delay between the Analog Input of the Headset and the Digital Input of the Base Station due to the Analog to Digital converter in the Headset and the wireless transmission between Headset and Base Station.

Due to the intrinsic filtering and down sampling of the ADC for analog channels, there is a delay between the analog biosignal input and digital trigger input lines (2 to 3 samples). For the electrode input signal a 70 % threshold was used for specify logic HIGH level.

Due to an internal ring buffer in the Headset, that is able to buffer 11 frames to minimize data loss, an additional delay (up to 11 samples) may be added, depending on the quality of the connection.

14.5 RF module

Frequency band	2.4 GHz
Transmission power	+3 dBm
IEEE standard	802.15.4
Marking	CE, ETSI, FCC, ISED
FCC ID:	XVV-MEGA22M00
Contains IC:	8720A-22M00

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

14.6 Maximum voltages at the following sockets on Base Station

USB	5 V DC
DIGITAL IN	5 V DC

14.7 Digital inputs

Sensitivity:	3.3 V / 5 V
Logic Low:	$\leq 0.66 \text{ V}$
Logic High:	$\geq 2.31 \text{ V}$

15 Electromagnetic compatibility g.Nautilus PRO

Please keep in mind the precautions in this Instructions for Use manual before installing and operating g.Nautilus. Note that HF-communication devices (e.g. mobile phones, routers etc.) may interfere with electric devices. g.Nautilus must not be used near, or stockpiled with, other devices. Only the original components for g.Nautilus from g.tec medical engineering GmbH are to be used for this device. Using third party manufacturer accessories may result in increased emission or decreased functional immunity of g.Nautilus, and/or other problems. As electric and magnetic fields may interfere with the functional reliability of the device, avoid using g.Nautilus close to devices emitting electro-magnetic fields.

The following values are the values given by EN 60601-1-2 for electromagnetic immunity testing:

Immunity test	IEC 60601 test level	Compliance level
Conducted RF IEC 61000-4-6	3 Veff 150 kHz to 80 MHz	1 → $V1$ in V ⁷
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2,5 GHz	1 → $E1$ in V/m ⁷

During all tests, g.Nautilus PRO was in acquisition mode, which is the most significant operating mode for the wireless transmission.

15.1 Guidance and manufacturer's declaration – electromagnetic emission

The g.Nautilus is intended for use in the electromagnetic environment specified below.

The customer or the user of the g.Nautilus should assure that it is used in such an environment.

Emission test	Compliance	Electromagnetic environment - guidance
RF emissions CISPR 11	Group 1	The g.Nautilus uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class B	
Harmonic emissions IEC 61000-3-2	Not applicable	The g.Nautilus is suitable for use in all establishments, including domestic establishments and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Voltage fluctuations/flicker emissions IEC 61000-3-3	Not applicable	

⁷ The lower compliance levels are allowed (according to normative EN60601-1-2, top 36.202.1 A) as g.Nautilus must be able to accurately measure biosignals with very low amplitudes. Using electromagnetic interference suppression would yield an inappropriately low signal to noise ratio for g.Nautilus.

15.2 Guidance and manufacturer's declaration – electromagnetic immunity

The g.Nautilus is intended for use in the electromagnetic environment specified below.

The customer or the user of the g.Nautilus should assure that it is used in such an environment.

Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment - guidance
Electrostatic discharge IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30 %.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	Not applicable as all cables have a maximum length of 3 meters.	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	± 1 kV differential mode ± 2 kV common mode	Not applicable as all cables have a maximum length of 3 meters.	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	< 5 % UT (> 95 % dip in UT) for ½ cycle 40 % UT (60 % dip in UT) for 5 cycles 70 % UT (30 % dip in UT) for 25 cycles < 5 % UT (> 95 % dip in UT) for 5 s	Not applicable as no mains supply port available.	Mains power quality should be that of a typical commercial or hospital environment. If the use of the g.Nautilus PRO requires continued operation during power mains interrupts, it is recommended that the g.Nautilus PRO be powered from an uninterruptible power supply or a battery.
Power frequency (50 Hz/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.



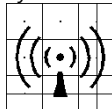
NOTE

UT is the a.c. mains voltage prior to application of the test level.

15.3 Guidance and manufacturer's declaration – electromagnetic immunity

The g.Nautilus is intended for use in the electromagnetic environment specified below.

The customer or the user of the g.Nautilus should assure that it is used in such an environment.

Immunity test	IEC 60601 test level	Compliance level	Electromagnetic environment - guidance
			Portable and mobile RF communications equipment should be used no closer to any part of the g.Nautilus PRO , including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter: Recommended separation distance
Conducted RF IEC 61000-4-6	3 V _{eff} 150 kHz to 80 MHz	3 → V ₁ in V *)	$d = \left(\frac{3,5}{V_1} \right) * \sqrt{P}$
Radiated RF IEC 61000-4-3	3 V/m 80 MHz to 2,5 GHz	3 → E ₁ in V/m *)	$d = \left(\frac{3,5}{E_1} \right) * \sqrt{P}$ 80 MHz to 800 MHz
		*) In the frequency range 20 MHz up to 400 MHz, the compliance level is only 1 V_{eff} or 1 V/m, respectively.	$d = \left(\frac{7}{E_1} \right) * \sqrt{P}$ 800 MHz to 2,5 GHz
			Where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m).^{b)} Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,^{a)} should be less than the compliance level in each frequency range.^{b)} Interference may occur in the vicinity of equipment marked with the following symbol: 

- Field strengths from fixed transmitters, such as base stations for radio (cellular /cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due

to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the g.Nautilus PRO is used exceeds the applicable RF compliance level above, the g.Nautilus PRO should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as reorienting or relocating the g.Nautilus PRO.

- Over the frequency range 150 kHz to 80 MHz, field strengths should be less than [V1] V/m.



NOTE

At 80 MHz and 800 MHz, the higher frequency range applies.



NOTE

These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

15.4 Recommended separation distances between portable and mobile RF communications equipment and the g.Nautilus PRO

The g.Nautilus PRO is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the g.Nautilus PRO can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the g.Nautilus PRO as recommended below, according to the maximum output power of the communications equipment.

Rated maximum output power of transmitter (W)	Separation distance according to frequency of transmitter (m)				
	150 kHz to 20 MHz	20 MHz to 80 MHz	80 MHz to 400 MHz	400 MHz to 800 MHz	800 MHz to 2,5 GHz
	$d = \left(\frac{3,5}{V1}\right) * \sqrt{P}$	$d = \left(\frac{3,5}{V1}\right) * \sqrt{P}$	$d = \left(\frac{3,5}{E1}\right) * \sqrt{P}$	$d = \left(\frac{3,5}{E1}\right) * \sqrt{P}$	$d = \left(\frac{7}{E1}\right) * \sqrt{P}$
0,01	0,12	0,35	0,35	0,12	0,23
0,1	0,37	1,11	1,11	0,37	0,74
1	1,17	3,50	3,50	1,17	2,33
10	3,69	11,07	11,07	3,69	7,38
100	11,67	35,00	35,00	11,67	23,33

For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be determined using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.



NOTE

At 80 MHz and 800 MHz, the higher frequency range applies.

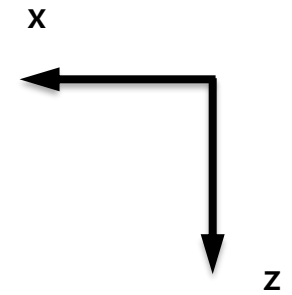
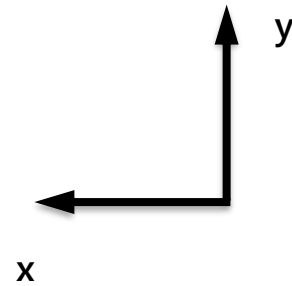


NOTE

These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.

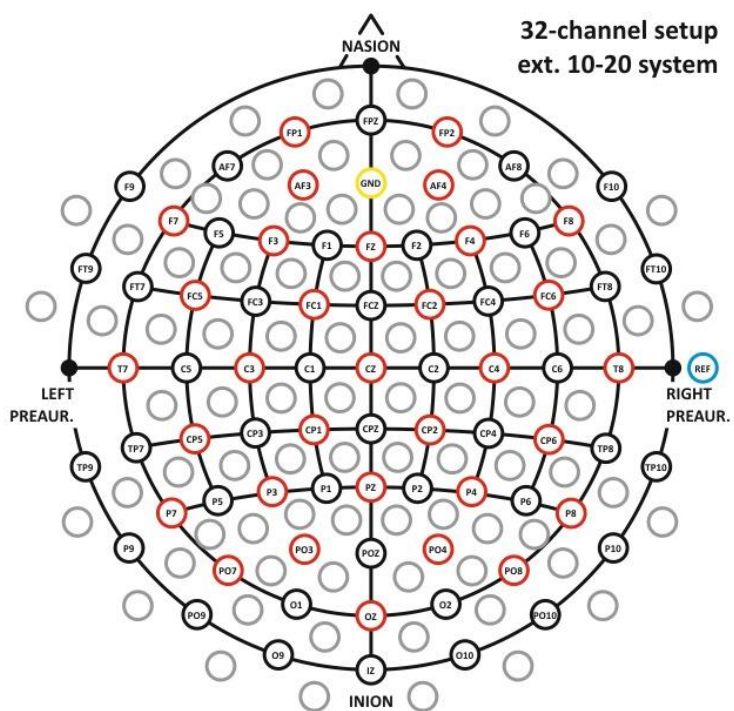
16 Appendix

16.1 Acceleration sensor



16.2 g.Nautilus PRO (32-channel setup)

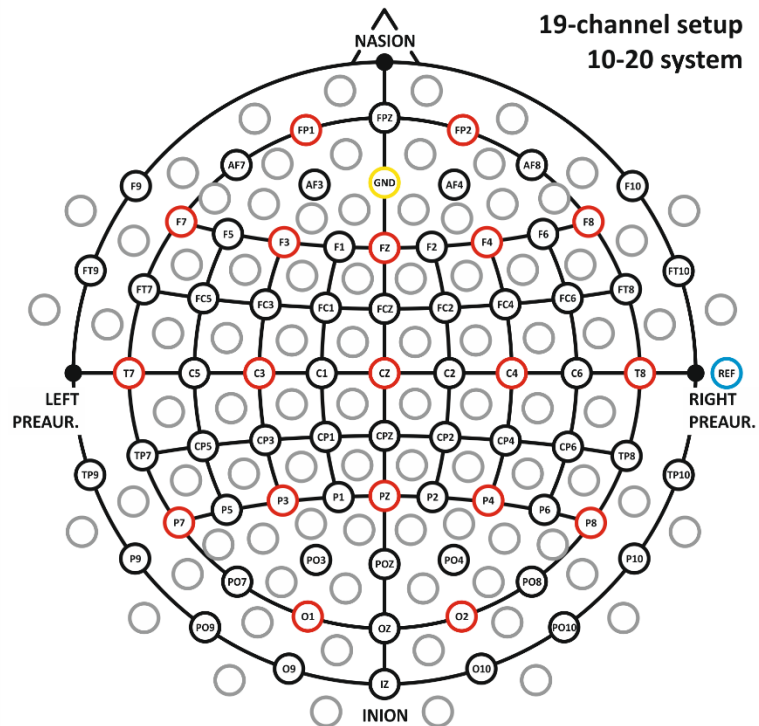
Electrodes	Logical Channel
FP1	1
FP2	2
AF3	3
AF4	4
F7	5
F3	6
FZ	7
F4	8
F8	9
FC5	10
FC1	11
FC2	12
FC6	13
T7	14
C3	15
CZ	16 ^{*)}
C4	17
T8	18
CP5	19
CP1	20
CP2	21
CP6	22
P7	23
P3	24
PZ	25
P4	26
P8	27
PO7	28
PO3	29
PO4	30
PO8	31
OZ	32



^{*)} Z-Channel (important for impedance measurement)

16.3 g.Nautilus PRO (19-channel setup)

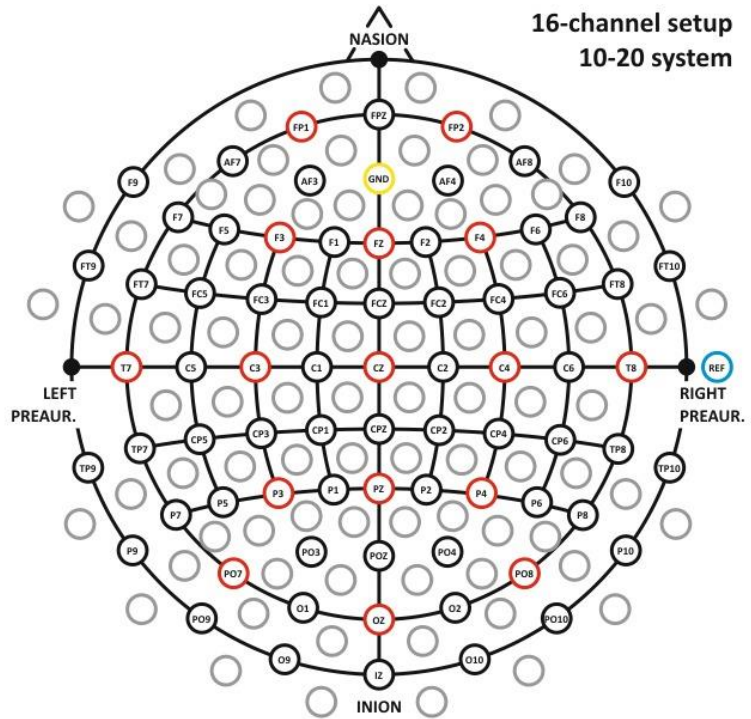
Electrodes	Logical Channel
FP1	1
FP2	2
F7	3
F3	4
FZ	5
F4	6
F8	7
T7	8
C3	9
CZ	10 ^{*)}
C4	11
T8	12
P7	13
P3	14
PZ	15
P4	16
P8	17
O1	18
O2	19



^{*)} Z-Channel (important for impedance measurement)

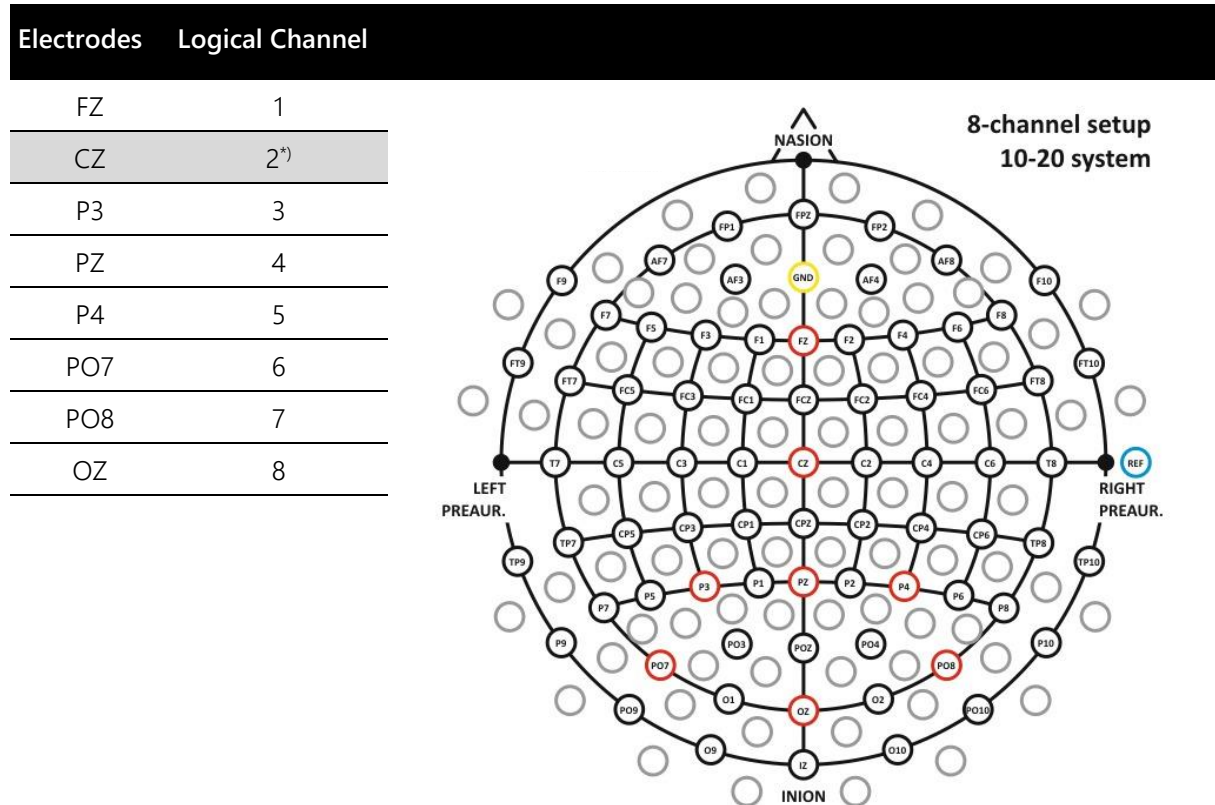
16.4 g.Nautilus PRO (16-channel setup)

Electrodes	Logical Channel
FP1	1
FP2	2
F3	3
FZ	4
F4	5
T7	6
C3	7
CZ	8 ^{*)}
C4	9
T8	10
P3	11
PZ	12
P4	13
PO7	14
PO8	15
OZ	16



^{*)} Z-Channel (important for impedance measurement)

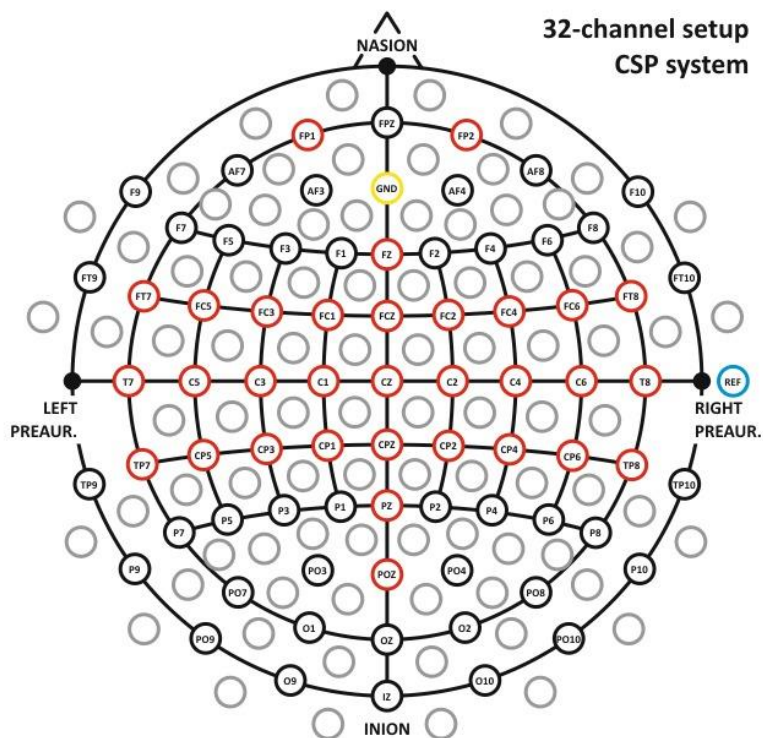
16.5 g.Nautilus PRO (8-channel setup)



^{*)} Z-Channel (important for impedance measurement)

16.6 g.Nautilus PRO (32-channel setup CSP system)

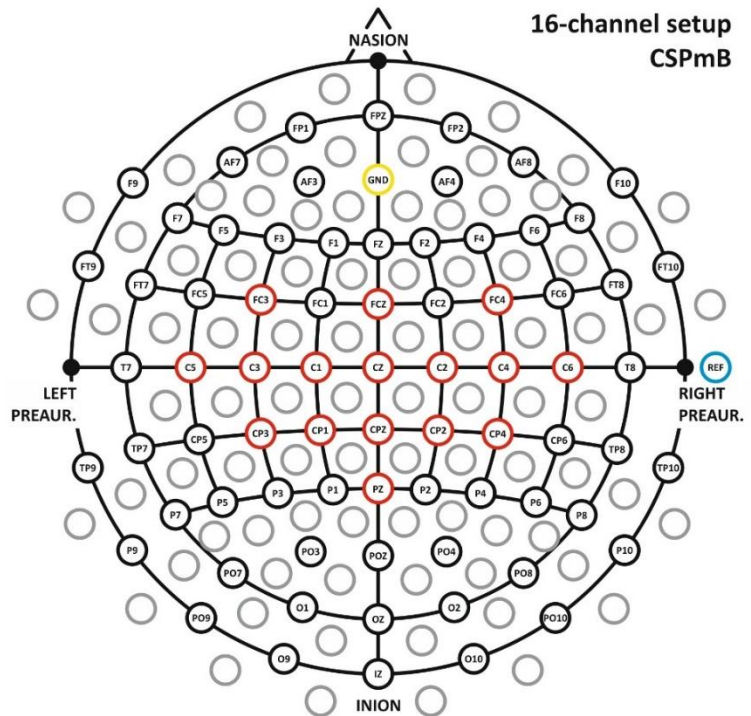
Electrodes	Logical Channel
FP1	1
FP2	2
FZ	3
FT7	4
FC5	5
FC3	6
FC1	7
FCZ	8
FC2	9
FC4	10
FC6	11
FT8	12
T7	13
C5	14
C3	15
C1	16
CZ	17 ^{*)}
C2	18
C4	19
C6	20
T8	21
TP7	22
CP5	23
CP3	24
CP1	25
CPZ	26
CP2	27
CP4	28
CP6	29
TP8	30
PZ	31
POZ	32



^{*)} Z-Channel (important for impedance measurement)

16.7 g.Nautilus PRO (16-channel setup CSPmB system)

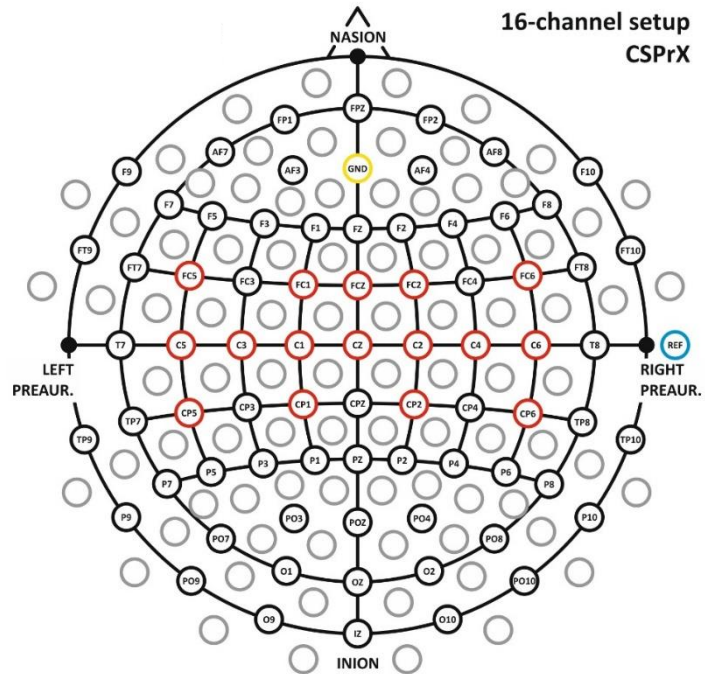
Electrodes	Logical Channel
FC3	1
FCZ	2
FC4	3
C5	4
C3	5
C1	6
CZ	7 ^{*)}
C2	8
C4	9
C6	10
CP3	11
CP1	12
CPz	13
CP2	14
CP4	15
Pz	16



^{*)} Z-Channel (important for impedance measurement)

16.8 g.Nautilus PRO (16-channel setup CSPrX system)

Electrodes	Logical Channel
FC5	1
FC1	2
FCZ	3
FC2	4
FC6	5
C5	6
C3	7
C1	8
CZ	9 ^{*)}
C2	10
C4	11
C6	12
CP5	13
CP1	14
CP2	15
CP6	16



^{*)} Z-Channel (important for impedance measurement)

16.9 g.Nautilus PRO Base Station pin assignment for digital inputs

Pin assignment for the 26 pin Sub D plug

Pin 1	digital input 1 (3.3 V tolerant)
Pin 2	digital input 2 (3.3 V tolerant)
Pin 3	digital input 3 (3.3 V tolerant)
Pin 4	digital input 4 (3.3 V tolerant)
Pin 5	digital input 5 (5 V tolerant)
Pin 6	digital input 6 (5 V tolerant)
Pin 7	digital input 7 (5 V tolerant)
Pin 8	digital input 8 (5 V tolerant)
Pin 9	NC
Pin 10	NC
Pin 11	NC
Pin 12	NC
Pin 13	NC
Pin 14	NC
Pin 15	NC
Pin 16	NC
Pin 17	+ 3.3 V dc
Pin 18	+ 3.3 V dc
Pin 19	GND
Pin 20	GND
Pin 21	NC
Pin 22	NC
Pin 23	NC
Pin 24	NC
Pin 25	NC
Pin 26	NC



16.10 Information to aid in the detection of failures

Item	Issue	Reason	Solution
1	Data loss during acquisition (indicated by Validation Indicator)	The chosen network channel is occupied or disturbed by another wireless device (WIFI, Bluetooth, ZigBee or another device using the 2.4 GHz ISM band)	- Choose another channel via end user software, or power cycle Headset and Base Station to initiate a new automatic channel selection - Eliminate interference sources such as wireless routers, mobile phones or Bluetooth devices
2	Data loss during acquisition (indicated by Validation Indicator)	Headset and its Base Station are out of range	Reposition Headset to be in wireless range of its Base Station (use Link Quality Indicator as measure).
3	No data is received (indicated by Validation Indicator)	Loss of battery power on Headset	Place Headset on charging pad
4	Error during configuration process	The chosen network channel is occupied by another wireless device (WIFI, Bluetooth, ZigBee or another device using the 2.4 GHz ISM band)	Power cycle Headset and Base Station to initiate new automatic channel selection
5	Headset and/or Base Station indicated loss of pairing and do not automatically re-pair	The chosen network channel is occupied by another wireless device (WIFI, Bluetooth, ZigBee or another device using the 2.4 GHz ISM band)	Power cycle Headset and Base Station to initiate automatic pairing mechanism
6	Bad signal quality indicated in end user software	- Distance between Headset and its Base Station is too great - Adverse arrangement of Headset and its Base Station (dependent on spatial conditions)	- Reduce distance between Headset and its Base Station - Change spatial arrangement of Headset and its Base Station
7	Headset is in acquisition mode but Base Station is not	- Acquisition has been stopped while Headset was out of range, or wireless channel encountered heavy interference while acquisition was stopped - USB cable has been removed from Base Station while acquisition was running	- Turn off Headset - Wait until Base Station indicates non-paired state - Turn on Headset

8	Base Station is in acquisition mode but Headset is not	- Headset has been turned off while acquisition was running. - Headset has been placed on charging pad while acquisition was running. - Headset ran out of battery power - Wireless channel encountered heavy interference while acquisition was stopped	- Disconnect and reconnect USB cable from Base Station to initiate automatic pairing sequence. - Turn on Headset - Charge Headset
9	Headset is not being charged, although it has been placed on charging pad.	- Headset is not centered on charging pad. - Headset has no direct contact with the charging pad (something is placed between charging pad and Headset, e.g. electrode cap). - Headset is not placed on the correct side of the charging pad. - Charging pad is not correctly powered.	- Check positioning of Headset on charging pad. - Make sure that nothing is positioned between Headset and charging pad. - Place the Headset on the side with no labeling. - Check if USB cable of the charging pad is correctly connected with the charging pad, and the light turns green on the power adapter and the charging pad.
10	One or more electrodes do not provide a correct signal	- Bad electrode/skin contact - When using wet electrodes: - No gel in electrode - Gel has no contact with skin - When using dry electrodes: - Too much offset voltage	- Check electrode/skin contact - Apply gel to wet electrodes - Check condition of the electrodes' connection cables (look for damage) - Reduce input sensitivity with dry electrodes
11	Headset cannot be turned on or signalling LED is blinking quickly in orange	Battery is too low for correct operation	Place Headset on charging pad to start charging cycle
12	Impedance measurement is not working properly	- Distance between Headset and its Base Station is too high - The used wireless channel is highly disturbed	- Minimize distance between Headset and its Base Station - Change wireless channel
13	Base Station is not able to pair with its Headset Headset is not able to pair with its Base Station	- Headset or Base Station is not powered on - Wrong combination of Headset and Base Station is used.	- Power on Headset and Base Station - Confirm that the serial numbers of Headset and Base Station are equal

16.11 Manual Best Channel Selection

g.Nautilus PRO includes an automatic channel selection algorithm to determine and select the best possible wireless channel for wireless data transmission. The channel selection algorithm is executed as soon as the g.Nautilus PRO Base Station is powered on. The measurements executed by this algorithm are always only a snapshot of the actual situation within the transmission band. Hence, future data acquisition may be affected by changes in the situation within the transmission frequency band (e.g. interferers appear after channel selection algorithm is executed). If problems related to the wireless connection between g.Nautilus PRO Headset and Base Station occur, we recommend performing the following steps to manually select the optimal g.Nautilus PRO network channel and achieve optimal performance for measurement with g.Nautilus PRO.

Step 1: Eliminate all possible sources of interference.

Since g.Nautilus operates within the worldwide free usable 2.4 GHz ISM band, it is essential for optimal device operation that there are no possible interferers enabled that are operating within this frequency band. Therefore, please make sure that:

- If applicable, please make sure that all existing g.Nautilus PRO devices in the vicinity use their own channel separated by at least one channel, e.g. 11, 13, 15 etc., or are switched off when not used (Headsets and Base Stations)
- If applicable, please make sure that all Bluetooth devices (e.g. Smartphones) within the operating range of g.Nautilus PRO are disabled.
- If applicable, please make sure that all ZigBee devices within the operating range of g.Nautilus PRO are disabled.
- If applicable, please make sure that all wireless LAN devices within the operating range of g.Nautilus PRO are disabled.
- If applicable, please make sure that all other devices that are operating within the 2.4GHz ISM band are disabled.

Step 2: Check for existing wireless LAN networks.

Use a Wi-Fi scanning software to examine wireless LAN situation in vicinity of your g.Nautilus PRO device. Check the signal strength of all wireless LAN channels. Identify which Wi-Fi channels are used by the existing wireless LAN networks, and which networks have the lowest signal strengths.

Step 3: Connect the g.Nautilus PRO Base Station to your PC via the provided USB cable to initiate the automatic channel selection and the pairing process between g.Nautilus PRO Headset and Base Station.

Step 4: Turn on the g.Nautilus PRO Headset by pressing the power on button for approximately 3 seconds.

Step 5: As soon as g.Nautilus PRO Headset and Base Station have successfully paired (indicated by LED), check the chosen network channel.

Step 6: Use the following table to determine whether the channel actually used by g.Nautilus PRO overlaps with an occupied Wi-Fi channel examined in step 2.

g.Nautilus PRO Channel	Overlap with Wi-Fi Channel
11	1, 2
12	1, 2, 3
13	1, 2, 3, 4
14	1, 2, 3, 4, 5
15	2, 3, 4, 5, 6
16	3, 4, 5, 6, 7
17	4, 5, 6, 7, 8
18	5, 6, 7, 8, 9
19	6, 7, 8, 9, 10
20	7, 8, 9, 10, 11
21	8, 9, 10, 11, 12
22	9, 10, 11, 12, 13
23	10, 11, 12, 13
24	11, 12, 13
25	12, 13, 14

Step 7: Via g.Nautilus PRO end user software, manually select a network channel. Apply the following rules to select the channel:

- If more than one g.Nautilus PRO channel exists that does not overlap with an existing Wi-Fi channel, choose the one furthest away from this Wi-Fi channel.
- If only one g.Nautilus PRO channel exists that does not overlap with an existing Wi-Fi channel, choose this one.
- If no g.Nautilus PRO channel exists that does not overlap with an existing Wi-Fi channel, choose one that overlaps with the Wi-Fi channel with the lowest signal strength.

Step 8: Verify performance of the wireless transmission by starting data acquisition and monitoring the validation indicator and the link quality indicator of the acquired data.

g.NAUTILUSPRO

WIRELESS BIOSIGNAL ACQUISITION